



Software Release Notes for UnityCloud

Release: UC-5.2.0
Date: May 2025

Affirmed Networks Inc.

A Microsoft Company

Copyright© 2025 Affirmed Networks, Inc. All Rights Reserved.

AFFIRMED & DESIGN®, ACUITAS SERVICE MANAGEMENT SYSTEM®, AFFIRMED NETWORKS MOBILE CONTENT CLOUD™, AFFIRMED OPEN WORKFLOW™, AFFIRMED MOBILE CONTENT CLOUD™, UnityCloud™, UnityCloud PaaS™, UnityCloud Platform™, UnityCloud Operations™, UnityCloud Ops™, Operations and Policy Manager (OPM)™ and other trademarks and designs are the registered or unregistered trademarks of Affirmed Networks, Inc. and its subsidiaries in the United States and in foreign countries. All other trademarks are the property of their respective owners. The Affirmed Networks, Inc. trademarks may not be used in connection with any product or service that is not Affirmed Networks' in any manner that is likely to cause confusion among customers or in any manner that disparages or discredits Affirmed Networks, Inc.

This document contains information that is the property of Affirmed Networks, Inc. This document may not be copied, reproduced, reduced to any electronic medium or machine readable form, or otherwise duplicated, and the information herein may not be used, disseminated or otherwise disclosed, except with the prior written consent of Affirmed Networks, Inc.

Contents

Release Abstract	9
How to Obtain Product Documentation.....	9
Customer Support	9
Virtual Machine Resource Requirements	10
UnityCloud Deployment Order.....	11
Supported OPM and Platform Images	12
Supported NFVI	13
Third-Party Software Versions.....	15
VM and VM ISSU Information	19
Openstack Heat Template changes.....	19
VM-OPM Openstack heat templates update values	19
K8s upgrade script	19
Misconfigured Openstack heat template features	20
AON updates for 5.2.0.....	21
K8s supported version	21
New Features and Enhancements	22
OPM Features.....	22
ADO-868232 Remove use of Mongo DB creds storage as identified by crypto board	22
ADO-1170282 OPM ISSU allows user configuration of helm options.....	22
ADO-1534827 In-service rotation of OPM MNOPM secrets with HashiCorp vault.....	22
SMF Features	23
ADO-1031811 SMF reject PDU session upon UPF failure/overload/UE pool exhaustion.....	23
ADO-1031813 Support single DNN with NI only	23
ADO-1031817 Virtual DNN based on vrf and charging characteristics	24
ADO-1031820 DN-AAA Radius UE IP assignment.....	24
ADO-1031835 Support SEPP discovery from NRF and SEPP redundancy.....	24
ADO-1540188 LI Security- LI key cannot be helm variable with clear text	24
UPF Features.....	25
ADO-620114 Support for UE Hair pinning in UC UPF (with NAT).....	25
ADO-1031858 PFD syntax backward compatibility.....	25

ADO-1063858 Support for SIP ALG.....	27
ADO-1329627 Change chunk sizing logic (large UE pools) when configured as zero.....	27
ADO-1550326 CGNAT security – Phase 1	28
ADO-1620869 Config checks for UE Pools, NAT Pools and Loopbacks.....	31
ADO-835463 Support of 1K VRFs and 3K DNNs	31
Platform and Infrastructure Features	31
ADO-1558434 Ability to provision individual or NESA mechid accounts for PaaS user interfaces Halo-e Integration in PaaS.....	31
ADO-1558449 SHA256-4096 and Higher PKI Key Size	32
ADO-1078164 Miscellaneous Routing – Netconf support for ribmgr and bgpMgr CLI	33
ADO-734007 Decoupling OPM from SMF/UPF	33
ADO-1741621 Transition from Banzai cloud kafka to Strimzi kafka	33
ADO-1040968 Support Show CLI pipelining for read-only users.....	33
ADO- 733992 Configuration Manager: Support for Secrets management from Hashicorp Vault.....	34
Unsupported and Tech Preview Features.....	35
Unsupported Features	35
Platform and Infrastructure	35
AMF	35
Tech Preview Features	35
Restrictions and Special Considerations.....	36
Upgrade Notes for UC5.2.0	37
ADO-208650 - Performance suite failure for increase in IPPE pod CPU usage.....	38
ADO-2070202 – Alerta and related charts not used/required	38
ADO-1980410 - SMF 5.1 Deployment Issue: Some fed-smf Pods Not Fully Ready After SMF 5.2 Deletion.....	38
ADO-2052828 – Rollback not completed	38
Config Changes	40
SMF Features	40
ADO-1031811 SMF reject PDU session upon UPF failures/UE pool exhaustion.....	40
ADO-1031813 Support single DNN with NI only	42
ADO-1031817 Virtual DNN based on vrf and charging characteristics	42
ADO-1031835 Support SEPP discovery from NRF and SEPP redundancy.....	44
ADO-1031820 DN-AAA Radius UE IP assignment.....	44

ADO-1540188 LI Security LI key cannot be helm variable with clear text	45
UPF Features	46
ADO-620114 UE Hairpin Support	46
ADO-1031858 PFD Syntax backward compatibility	46
ADO-1063858 Support for SIP ALG	47
ADO-1329627 Change chunk sizing logic (large UE pools) when configured as zero	47
ADO-1550326 CGNAT security – Phase 1	47
ADO-1620869 Config checks for UE Pools, NAT Pools and Loopbacks	49
ADO-835463 Support of 1K VRFs and 3K DNNs	51
ADO-1944155 UPF - VPP crash during call model ramp up	51
PaaS Features	52
ADO-1270324 Post-recovery pod rebalancing	52
ADO-1558434 Ability to provision individual or NESA mechid accounts for PaaS user interfaces Halo-e Integration in PaaS	56
Elastic Upgrade	57
Infra Features	57
ADO-734007 Decouple OPM from SMF-UPF	57
ADO-1558449 Support SHA256-4096 PKI Key Size	58
ADO-733995 Support Halo authentication for PaaS UIs	59
ADO-1741621 Transition from Banzai cloud kafka to Strimzi kafka	60
OPM Features	64
ADO-868232 OPM changes to remove use of Mongo DB creds storage as identified by crypto board	64
ADO-734007 Decouple OPM & AOSM from SMF-UPF	64
ADO-1534827 In-service rotation of Secrets with HashiCorp Vault- for both Network Cloud and Nexus (Non LI NFs and Modules)	64
Other Changes	65
Anomalies	67
New Anomalies in this release	67
SMF	67
OPM	68
Platform and Infrastructure	71
UPF	73
Existing Anomalies	74
AMF	75

Common Microservices	75
DPI	76
N3IWF	76
NRF	77
OPM	77
Platform and Infrastructure	78
SMF	82
UPF	82
Anomalies Fixed	85
Anomalies fixed in UC5.2.0	85
OPM	85
Platform and Infrastructure	86
SMF	86
UPF	87
NRF	88
Anomalies fixed in UC5.2.0.x	89

List of Tables

Table 1.	Minimum VM requirements to deploy UnityCloud	10
Table 2.	Supported OPM and Platform VM images	12
Table 3.	OpenStack Configuration	13
Table 4.	KVM Configuration	14
Table 5.	Third-Party Software Versions used in UnityCloud	15
Table 6.	Tech Preview Features	35
Table 7.	Restrictions and Special Considerations	36
Table 8.	New Anomalies in SMF	67
Table 9.	New Anomalies in OPM	68
Table 10.	New Anomalies in Platform and Infrastructure	71
Table 11.	New Anomalies in UPF	73
Table 12.	Existing Anomalies in AMF	75
Table 13.	Existing Anomalies in Common Microservices	75
Table 14.	Existing Anomalies in Lawful Intercept (LI)	76
Table 15.	Existing Anomalies in DPI	76
Table 16.	Existing Anomalies in N3IWF	76
Table 17.	Existing Anomalies in NRF	77
Table 18.	Existing Anomalies in OPM	77
Table 19.	Existing Anomalies in Platform and Infrastructure	78
Table 20.	Existing Anomalies in SMF	82
Table 21.	Existing Anomalies in UPF	82
Table 22.	Anomalies Corrected in OPM	85
Table 23.	Anomalies Corrected in Platform and Infrastructure	86
Table 24.	Anomalies Corrected in SMF	87
Table 25.	Anomalies Corrected in UPF	87

Table 26. Anomalies Corrected in NRF 88

Release Abstract

This document contains the software release notes for the Affirmed Networks UnityCloud current release.

This release has been produced in accordance with Microsoft's Secure Development Lifecycle, including processes for authorizing software changes, antimalware scanning, and scanning and mitigating security bugs and vulnerabilities.

How to Obtain Product Documentation

Documentation is available for currently supported product releases. You can access Affirmed Networks Guides on the [Manuals](#). From here, you can also download a PDF version of an individual 2048guide, or the full manual set for a product. For access and instructions, please contact Support.

Customer Support

Affirmed Networks Support provides technical support 24 hours a day, 7 days a week (24 x 7) for any customer with an active maintenance contract.



E-mail Support: [Email support](#)
Open a Ticket: [Open a support ticket](#)



24 x 7 Telephone support:
Local: 1-978-268-0871
Toll free: 1-888-283-2838

Virtual Machine Resource Requirements

The following table provides the minimum Virtual Machine (VM) requirements for deploying UnityCloud VMs in this release.

Note: For the Operations and Policy Manager (OPM) VM, the minimum disk space is 25G. However, if you plan to use OPM's Docker Registry and Helm Chart Repository, you will need enough disk space to hold all the container images and charts.

Table 1. Minimum VM requirements to deploy UnityCloud

VM	IP Address	vCPU	Memory	Disk Needed if VM ISSU disabled	Disk Needed if VM ISSU enabled
OPM	10.165.5.10	4	16G	200G	400G
Master-1	10.165.5.11	4	8G	150G	300G
Node-1	10.165.5.12	16	78G	150G	300G
Node-2	10.165.5.13	16	78G	150G	300G
Node-3	10.165.5.14	16	78G	150G	300G
Node-4	10.165.5.15	16	78G	150G	300G
Node-5	10.165.5.16	16	78G	150G	300G
Node-6	10.165.5.17	16	78G	150G	300G
Node-7	10.165.5.18	16	78G	150G	300G
Node-8	10.165.5.19	16	78G	150G	300G
Node-9	10.165.5.20	16	78G	150G	300G
Node-10	10.165.5.21	16	78G	150G	300G
Node-11	10.165.5.22	16	78G	150G	300G
Node-12	10.165.5.23	16	78G	150G	300G
Node-13	10.165.5.24	16	78G	150G	300G

UnityCloud Deployment Order

For the Deployment Order information, see the *UnityCloud Deployment Guide*.

Supported OPM and Platform Images

This release supports the following platform and OPM VM images.

These images are created at customer site. They are built from the following source tgz

- uc-vm-pipeA-vm-platform_split-5.2.0-2-rel-5-2-0.tgz
- uc-vm-pipeA-vm-opm_split-5.2.0-3-rel-5-2-0.tgz

Table 2. Supported OPM and Platform VM images

VM	Supported Images
Platform CM	• vm-platform_split-5.2.0-2-rel-5-2-0.qcow2
OPM VM	• vm-opm_split-5.2.0-3-rel-5-2-0.qcow2 • vm includes fed-crds-5.2.0-16-rel-5-2-0.tgz • vm includes fed-opm-5.2.0-19-rel-5-2-0.tgz • vm includes fed-rbac-5.2.0-9-rel-5-2-0.tgz • These Helm chart on the vm are located here /opt/affirmed/commissioning/templates

Supported NFVI

The following third-party hardware platforms and software versions are qualified by Affirmed Networks for use with this release. If you are running a different permutation, contact Support for guidance on whether that permutation is supported.

The following table lists the OpenStack configuration software requirements for each hardware platform.

Table 3. OpenStack Configuration

Hardware Platform	OpenStack Configuration					
	VIMs	Operating System	Switching Technology	NICs	Firmware Version	Host Driver
Dell R740	OpenStack Stein	Ubuntu 18.04.5 LTS	SR-IOV OVS	Intel XXV710 (Fortville) 25Gb/s	7.10 0x800075e6 19.5.12	i40e 2.12.6.3
	OpenStack Queens RHOSP 13.0.11	RHEL 7.7	OVS-DPDK	Mellanox ConnectX-4 25Gb/s	14.24.8000	mlx_core 5.0-0
Dell R740	OpenStack Train RHOSP 16.2	RHEL 8.4	SR-IOV OVS	Intel XXV710 (Fortville) 25Gb/s	7.10 0x800075e6 19.5.12	i40e 2.12.6.3
	OpenStack Queens RHOSP 13.0.11	RHEL 7.7	OVS-DPDK	Intel XXV710 (Fortville) 25Gb/s HPE Eth 10/25Gb 2p 640FLR-SFP28 ConnectX-4	7.10 0x800075e6 19.5.12 14.26.1040	i40e 2.8.20-k mlx5_core 5.0-0

The following table lists the KVM configuration software requirements for each hardware platform:

Table 4. KVM Configuration

Hardware Platform	KVM Configuration					
	VIMs	Operating System	Switching Technology	NICs	Firmware Version	Host Driver
HPE DL380 Gen 10	virsh/Cockpit	RHEL 9.2	SR-IOV	817749-B21 (HPE Ethernet 25Gb 2-port 640FLR-SFP28 Adapter) 25G 817753-B21 (HPE Ethernet 25Gb 2-port 640SFP28 Adapter) 25G	14.31.1200 (HP_2420110034)	mlx5_core 23.04-1.1.3
HPE DL380 Gen 10 Plus	virsh/Cockpit	RHEL 9.2	SR-IOV	MLX MCX623106AS 100GbE 2p QSFP56 Adptr 100G MLX MCX631102 10/25GbE 2p SFP28 Adptr 100G	22.32.2004 (MT_0000000437)	mlx5_core 23.04-1.1.3

Third-Party Software Versions

The following table lists the PaaS components and other third-party software versions used in Third-Party Software Versions used in this release.

Table 5. Third-Party Software Versions used in UnityCloud

Category	Project/ Software	Version Used in this release	Link to Releases Page
CaaS	VM guest OS	RHEL 9.5 Kernel: 5.14.0- 503.26.1.el9_5.x86_64	
CaaS	Containerd	1.6.33	
CaaS	Kubernetes	1.30.9	
CaaS	coredns	1.11.3	
CaaS	K8s etcd	3.5.15-0	
CaaS	metrics-server	v0.7.2	
CaaS	openstack- cloud- controller- manager	v1.24.2	
CaaS	k8s-dns-node- cache	1.22.23	
CaaS	csi-node- driver- registrar	v2.5.0	
CaaS	csi-resizer	v1.4.0	
CaaS	livenessprobe	v2.6.0	
CaaS	cinder-csi- plugin	v1.24.2	
CaaS	csi-attacher	v3.4.0	
CaaS	csi-provisioner	v3.1.0	
CaaS	csi- snapshotter	v5.0.1	
CaaS	Helm	3.16.4	https://github.com/helm/helm/releases
CaaS	Calico	3.28.2	https://github.com/projectcalico/calico/releases
CaaS	MetalLB	0.8.2	https://github.com/metallb/metallb/releases
CaaS	Multus CNI	3.1 (commit bf61002)	https://github.com/intel/multus-cni/releases

Third-Party Software Versions

Category	Project/ Software	Version Used in this release	Link to Releases Page
CaaS	SRIOV CNI	~2.0.0 (commit hash 9e4c973b2ac517c64867e 33d61aee 152d70dc330)	https://github.com/intel/sriov-cni/releases
CaaS	Docker Registry	2.6.2	https://hub.docker.com/_/registry?tab=tags
CaaS	Chart Museum	0.8.2	https://hub.docker.com/r/chartmuseum/chartmuseum/tags
CaaS/ IaaS	Ceph	18.2.1	https://docs.ceph.com/docs/master/releases/
CaaS/ IaaS	Rook	1.13.3	https://github.com/rook/rook/releases
CaaS/ IaaS	HAProxy	haproxy-2.4.22- 3.el9_3.x86_64	Releases · haproxy/haproxy (github.com)
CaaS/ IaaS	Keepalived	keepalived-2.2.8- 3.el9.x86_64	Releases · acassen/keepalived (github.com)
PaaS	Istio	OPERATOR_IMAGE = istio/operator:1.19.0.1 PILOT_IMAGE = istio/pilot:1.19.0.1 PROXY_IMAGE = istio/proxyv2:1.19.0.1 CNI_IMAGE = istio/install- cni:1.19.0.1	https://github.com/istio/istio/releases
PaaS	Cert-Manager	1.15.3	https://github.com/jetstack/cert-manager/releases
PaaS	Etcd	3.5.14	https://github.com/etcd-io/etcd/releases
PaaS	etcd- operator(in- house)	0.10.0	https://github.com/coreos/etcd-operator/releases
PaaS	Elastic	8.14.1	https://docs.search-guard.com/latest/search-guard-versions
PaaS	Curator	8.15	https://docs.search-guard.com/latest/search-guard-versions
PaaS	ElastAlert(in- house)	3.0.0-beta.1	https://github.com/bitsensor/elastalert/releases
PaaS	Fluentd	1.18.2	https://github.com/fluent/fluentd
PaaS	Fluentd Operator	1.18.2	https://github.com/vmware/kube-fluentd-operator

Third-Party Software Versions

Category	Project/ Software	Version Used in this release	Link to Releases Page
PaaS	KubeDB Operator	v2024.4.4	https://github.com/kubedb/operator
PaaS	Kibana	8.14.1	https://www.elastic.co/guide/en/kibana/current/release-notes.html
PaaS	Jaeger	1.59.0	https://github.com/jaegertracing/jaeger/releases
PaaS	Prometheus	2.54.1	https://github.com/prometheus/prometheus/releases
PaaS	Prometheus Operator	0.76.2	https://github.com/coreos/prometheus-operator/releases
PaaS	Grafana	11.2.3	https://github.com/grafana/grafana/releases
PaaS	AlertManager	0.27.0	https://github.com/prometheus/alertmanager/releases
PaaS	Consul	1.7.1	https://www.consul.io/downloads.html https://github.com/hashicorp/consul/releases
PaaS	Apache Fileserver	2.4.48	https://httpd.apache.org/
PaaS	Kafka	3.7.1	https://github.com/apache/kafka/releases
PaaS	Zookeeper	0.2.15	https://github.com/apache/zookeeper/releases
PaaS	Kafka operator	0.43.0	https://github.com/banzaicloud/kafka-operator/releases
PaaS	Zookeeper operator	0.2.15	https://github.com/pravega/zookeeper-operator/releases
PaaS	Kube- Prometheus	2.52.0	https://github.com/prometheus-operator/kube-prometheus/release
PaaS	OpenPolicy Agent (OPA)	0.54.0	https://github.com/open-policy-agent/opa/releases
PaaS	kube-mgmt	8.3.0	https://github.com/open-policy-agent/kube-mgmt/releases
PaaS	Thanos	0.34.1	https://github.com/thanos-io/thanos/releases
PaaS	prometheus- adapter	0.12.0	https://github.com/kubernetes-sigs/prometheus-adapter/releases
PaaS	node-exporter	1.4.0	https://github.com/prometheus/node_exporter/releases
PaaS	prometheus config reloader	0.74.0(mcr)	https://github.com/prometheus-operator/prometheus-operator/pkg/container/prometheus-config-reloader

Third-Party Software Versions

Category	Project/ Software	Version Used in this release	Link to Releases Page
PaaS	config map reload	0.8.0	https://github.com/jimmidyson/configmap-reload/tags
PaaS	local_path- provisioner	0.0.29	https://github.com/rancher/local-path-provisioner/releases
PaaS	Jmx- Javaagent	0.14.0	https://github.com/banzaicloud/jmx_exporter/tags
PaaS	oauth2-proxy	7.7.1	https://github.com/oauth2-proxy/oauth2-proxy
PaaS	Kafka Cruise control	2.5.133	https://github.com/banzaicloud/docker-cruise-control/tags
App	Libcurl	7.80.0 (Plus fixes)	https://github.com/curl/curl/releases
App	libmicrohttpd	0.9.72 (Plus HTTP/2 stack)	https://ftp.gnu.org/gnu/libmicrohttpd/
App	nghttp2	1.46.0	https://github.com/nghttp2/nghttp2/releases
App	opentracing- client-cpp	1.5.0.1	https://github.com/opentracing/opentracing-cpp/releases
App	jaeger-client- cpp	0.4.2	https://github.com/jaegertracing/jaeger-client-cpp/releases
App	Mongo DB C Client	1.22.0 (rel_3_2_0, rel3_3_0)	https://github.com/mongodb/mongo-c-driver
App	Mongo DB Go Client	1.4.4	https://github.com/mongodb/mongo-go-driver
App	Prometheus Client	0.12.1	https://github.com/jupp0r/prometheus-cpp/releases
App	Redis	7.0.12	https://github.com/redis/redis/releases

VM and VM ISSU Information

Deploying this version of the VM will create Kubernetes version **1.30.9 cluster**.

Openstack Heat Template changes

ff_k8sversion

values(strings):

- **1.29.13**
 - **Deploys K8s 1.29.13**
- **1.30.9 (default)**
 - **Deploys K8s 1.30.9**

***Any other value than 1.29.13 or 1.30.9 will default to 1.30.9**

However, if this parameter is not in the Openstack heat template, then the default Kubernetes version will be 1.30.9

Heat template for vm-platform should include **ff_configparser_override** parameter.

This is not new and has been part of the heat template since UC 3.1.

VM-OPM Openstack heat templates update values

- **opm_helm_chart:** /opt/affirmed/commissioning/templates/ fed-opm-5.2.0-19-rel-5-2-0.tgz
- **crds_helm_chart:** /opt/affirmed/commissioning/templates/ fed-crds-5.2.0-16-rel-5-2-0.tgz
- **rbac_helm_chart:** /opt/affirmed/commissioning/templates/ fed-rbac-5.2.0-9-rel-5-2-0.tgz

K8s upgrade script

If manually running vm-issu,, run this cmd on each Masters VM once the whole cluster has been upgraded. **/opt/affirmed/scripts/ucK8sUpgrade.py**

Misconfigured Openstack heat template features

1. Allow ucUpgrader.py to ignore case where customer doesn't have ff_configparser_override included in their Openstack heat template.
2. Allow configParser.py to check if dreg.crt was added to vm-platform manually and not added via Openstack heat template **dreg_cert** parameter.

AON updates for 5.2.0

This section provides information regarding updates related to UC 5.2.0 on AOM.

K8s supported version

Both SMF and UPF installed and validated on:

- NAKS k8s version 1.30.5
- AON NC version - 3.14.3
- AON NNF Version - 4.0.0
- AoSM NFO - 2.0.2875-165

New Features and Enhancements

This section lists the new features and enhancements introduced in this release. For information regarding config changes per feature, see [Config Changes per Feature](#).

For information on features and enhancements included in previous releases, refer to the respective release notes.

OPM Features

ADO-868232

Remove use of Mongo DB creds storage as identified by crypto board

This feature addresses the following crypto ticket:

- Remove Mongo Db storage of user credentials that are encrypted using BCrypt

Consequently, all OPM credentials in Mongo or file system are moved to Kubernetes secrets. Note Kubernetes secret values are encoded as base64 strings and are stored unencrypted by default but are recommended to be encrypted at rest to improve security. Alternatively, we recommend storing sensitive data in external vault such as Hashicorp vault for best protection.

ADO-1170282 OPM ISSU allows user configuration of helm options

This feature allows users to configure helm options, such as wait, debug, timeout, etc. that are used to execute helm upgrade and test operations per federation during ISSU.

ADO-1534827

In-service rotation of OPM MNOPM secrets with HashiCorp vault

The integration of HashiCorp Vault with MNOPM (Multi node OPM) federation is a strategic initiative aimed at enhancing the security and management of sensitive information within the Unity Cloud solution from UC 5.2 release. This integration focuses on migrating sensitive data such as usernames, passwords and connection strings from local storage to the HashiCorp Vault, ensuring secure and centralized management of secrets.

Key Features

- **Vault Configuration:** Helm specific value JSON allowing users to enable or disable the HashiCorp Vault, set the server URL, define roles, and specify authentication methods.

- **Sensitive Data Management:** All sensitive data used in the fed-opm_multi (multi-node) deployment, including credentials will be stored in the HashiCorp Vault. This eliminates plaintext sensitive data from Helm values and replaces them with vault path references.
- **Automatic Secret Rotation:** OPM will manage the rotation of secrets configured in the Vault server automatically, reducing the need for manual intervention.
- **Exclusions:** Single Node OPM is excluded from this integration due to its lightweight nature and the impracticality of using Kube add-ons and a running K8 cluster

To ensure a smooth integration of HashiCorp Vault with OPM, the following pre-requisites must be met:

- **Vault Server Management:** Each customer will manage their own HashiCorp Vault server.
- **Centralized Vault Server:** A centralized (external) Vault server will be used, with multiple Kubernetes clusters interacting with it.
- **Authentication Method:** JWT authentication method with static keys is preferred.
- **Secret Setup:** OPM secret information must be set up in the Vault before deployment.
- **Vault Availability and Configuration:** The availability and configuration of the Vault server will be managed by the customer's HashiCorp Vault team.

SMF Features

ADO-1031811

SMF reject PDU session upon UPF failure/overload/UE pool exhaustion

This feature supports the early rejection of PDU session setup when various UPF failures are encountered, such as N4 failure (none of the configured PFCP associations to UPF are Up) and UE pool exhaustion. With this feature, the SMF receives DNN Local UE IP Pool Utilization information from the UPF, which enables the SMF to make an early decision on the session setup.

ADO-1031813

Support single DNN with NI only

This feature allows any NI+OI incoming DNN to be matched to a configured NI-only DNN. This eliminates the need to configure multiple DNNs with the same NI part.

ADO-1031817

Virtual DNN based on vrf and charging characteristics

This feature lets operators use SMF analyzer configuration to map a UE-requested DNN to a virtual DNN, allowing different behavior for UEs with the same DNN and properties (IMSI, PLMN, Charging-Characteristics etc.).

ADO-1031820

DN-AAA Radius UE IP assignment

This feature allows SMF to support IP assignment for the UE as part of Radius access-accept message. This enables enterprise customers to use this capability instead of the static IP solution to control the IP address to UE identity binding. This ensures customers have effective control of IP assignments using pools based on DNN and other parameters rather than individual IP address provisioning at per UE level vs the static IP solution. When radius authentication profile is configured in the management profile.

Configuration attribute `ue-ip-allocation-mode` is added to `smf service-construct subscriber-management-profile <mgmt.-profile>` allows end users the choice of local or Radius Server for UE IP address allocation `<[local (default value)] | radius>`.

ADO-1031835

Support SEPP discovery from NRF and SEPP redundancy

This feature supports SEPP discovery from NRF and SEPP redundancy when a request from SMF sent to one SEPP fails, the alternate SEPP will be tried for the same request.

ADO-1540188

LI Security- LI key cannot be helm variable with clear text

This feature supports having LI Encryption key in the Hashicorp Vault instead of configuring as helm variable in plain text format. From this release onwards, LI Encryption Key can be configured either as helm variable or can be fetched from Hashicorp vault.

Also, ISSU support is added for LI tasks, ISSU can be done from 5.2.0 to future releases with LI tasks active. The existing MOP steps will be effective for upgrading from version 5.1 to 5.2.

UPF Features

ADO-620114

Support for UE Hair pinning in UC UPF (with NAT)

This feature adds support for all possible intra-DNN Hairpin Traffic (UE-to-UE) cases. Hairpin Traffic occurs when two UEs, connected to the same UPF, communicate directly with each other.

New configuration is added to turn this feature on/off:

```
upf service-construct dnn <dnn_name>
  dnai-list                Data Network Access Identifier List name
  ipv4-ue-pool              IPv4 UE Pool to be associated with the DNN
  ipv6-ra-param
  ipv6-ue-pool              IPv6 UE Pool to be associated with the DN
  network-context          Network-Context to be associated with the DNN
  pdu-session-type         Type of UE-IP address supported for this DNN
  service-profile           Service profiles to be associated with the DNN
  allow-ue-to-ue            true/false
```

Statistics are added to count the:

- Hairpin number packets allowed
- Hairpin number packets disallowed due to config
- Hairpin number packets TX dropped no default route
- Hairpin number packets error dropped no NC with fib idx

ADO-1031858

PFD syntax backward compatibility

Packet Flow Description (PFD) is used to classify user traffic to apply certain actions on the user traffic. This feature addresses the need for subsequent releases of Unity Cloud to support the format of PFDs from previous Unity Cloud releases. The PFD baseline for this feature is Unity Cloud 4.2. With this feature, operators can display the status of the PFD installed on the UPF on a per PFD basis using operator CLI command. The reason for failure is also displayed. The main benefits of this feature are:

- Helps verify the status of dynamic PFDs installed on UPF, checks the reason for PFD failure if applicable
- Ensures PFDs supported in 4.2.0 are supported in subsequent releases

The show ppe v1 pfd-info command includes 'status' and 'failed-rules', see the following example:

```
admin@pod-cfgmgr-5b648476bd-6687w 12:49:37# show ppe v1 pfd info
```

New Features and Enhancements

```
{
  "ApplicationId": [
    {
      "application-identifier": "windows_ul_2",
      "pfds": [
        {
          "domain-name": "",
          "domain-name-protocol": "",
          "flow-descriptions": "permit in 6 from 10.10.62.0/24 1000 to any",
          "status": "Installed",
          "urls": ""
        }
      ]
    }
  ],
  "instanceId": "0",
  "ppeId": "ppe-548464544-jvcbm",
  "timestamp": "2024-11-02T12:54:14.749819"
}
```

admin@pod-cfgmgr-5b648476bd-6687w 12:49:37# show ppe v1 pfd-info

```
{
  "ApplicationId": [
    {
      "application-identifier": "app_id_1",
      "pfds": [
        {
          "domain-name": "",
          "domain-name-protocol": "",
          "failed-rules": [
            "permit any 6 from 10.10.62.0/24 80,443,22,21 to any 80"
          ],
          "flow-descriptions": "permit any 6 from 10.10.62.0/24 80,443,22,21 to any 80",
          "status": "Flow description validation failed",
          "urls": ""
        }
      ]
    }
  ],
  "instanceId": "0",
  "ppeId": "ppe-548464544-jvcbm",
  "timestamp": "2024-11-02T12:54:14.749819"
}
admin@pod-cfgmgr-5b648476bd-6687w 12:54:14#
```

ADO-1063858

Support for SIP ALG

With this feature, the UPF supports ALG parity with the MCC GW (Gateway) to cover IPv4 NAT and IPv6 PASS for SIP ALG.

This feature introduces parameter sip under alg.

```
upf service-construct firewall-service-policy FSP

priority 100

nat-pool NATPOOL

action nat

alg sip

!
```

The show ppe v1 firewall nat alg sip statistics <poid-id> command is updated to display the aggregated statistics across all workers.

ADO-1329627

Change chunk sizing logic (large UE pools) when configured as zero

This feature updates the chunking logic for pools when the chunksize is 0 (auto-calculate the chunksize). Existing CLI commands are enhanced and new CLI commands added to display the route count related information for committed configuration.

These commands are added:

```
show bgp-mgr v1 routecount ppe-pod-id <ppe-id>
```

This command displays the total number of routed advertised/accepted under each address-family in a PPE pod. The output includes the Primary/Secondary routes for the RBs - across UE Pools, NAT Pools, Loopbacks, Host routes etc.

```
show bgp-mgr v1 routecount ppe-pod-id <ppe-id> network-context <nc-name>
```

This command displays the total number of routed advertised/accepted under each address-family in a specific network-context in specific PPE pod. The output includes the Primary/Secondary routes for the RBs - across UE Pools, NAT Pools, Loopbacks, Host routes etc.

ADO-1550326

CGNAT security – Phase 1

This feature enables the following functionality:

- DDOS Mitigation:
 - A Distributed Denial of Service (DDoS) attack is a malicious attempt to disrupt the normal functioning of a targeted server, service, or network by overwhelming it with a flood of network traffic.
 - The purpose of this feature is to support mitigating the Distributed Denial of Service (DDoS) attack.
 - The feature will support the Protection from Bad IP and TCP/UDP header (malformed packet).
 - The feature will support the monitoring of connection establishment states, allowing for faster aging out in cases of half-opened connections.
 - Sample Configuration:
 - Existing config header-sanity-check is used to identify and drop the malformed packets if enabled.

```
upf service-construct network-context N3
tcp-mss-adjust-ipv4      0
tcp-mss-adjust-ipv6      0
type                     N3
header-sanity-check      true
ignore-notify-bgp-peer-state-change false
!
```

- By default, connection tracking is disabled for TCP/UDP half opened connections. User has to explicitly configure the tcp-flow-fast-age-timeout and udp-flow-fast-age-timeout

```
upf service-construct flow-idle-timer-profile FP
default-tcp-flow-idle-timeout      45
default-udp-flow-idle-timeout      30
default-non-tcp-udp-flow-idle-timeout 45
default-tvs-flow-idle-timeout      45
flow-idle-timer-rule fr
tcp-flow-idle-timeout              45
udp-flow-idle-timeout              30
non-tcp-udp-flow-idle-timeout      45
tvs-flow-idle-timeout              45
tcp-flow-fast-age-timeout          20
udp-flow-fast-age-timeout          25
```

!

!

- Flow Rate Limit

- This feature enables Flow Rate Limiting (FRL) functionality in the UPF. This FRL feature is used to control the rate of connections created in the UPF. FRL uses a simple custom policer that requires the connections rate to be configured. This feature helps protect the worker cores against a burst-of or flood-of new connection create flows.

Note: At present, FRL is supported at a global level and not at the NC or DNN level.

- Sample Configuration:

- Flow Rate Limit configuration is available as a service-construct profile CLI. If operators configure FRL service-construct, it is treated as global configuration and applied accordingly.
- The following is an example:

```
upf service-construct flow-rate-limit-profile abcd
new-flow-creation-rate-limit-ul 545
new-flow-creation-rate-limit-dl 1045
```

- Sample output:

- `admin@pod-cfgmgr-654d7dff79-6xk72 04:23:26# show ppe v1 vpp frl`

```
{
  "FlowRateLimitInfo":
  {
    "dl-cb-ms":0,
    "dl-cir-pps":100,
    "dl-policer-prev-color":"green",
    "dl-policer-state":1,
    "frlEnabled":true,
    "profile-name":"abcd",
    "ul-cb-ms":0,
    "ul-cir-pps":100,
    "ul-policer-prev-color":"red", "ul-policer-state":1
  }
}
```

- ACL Support

- ACLs (Access Control List) are used to control/manage network traffic using set of rules that specify which packets are allowed (or) denied in a network
- ACLs help in preventing malicious traffic from the network

- As a part of this requirement, the operator will have granular control over the traffic to be allowed/denied on a Network Context basis.
- As part of this feature, support is added in UPF to allow/deny packets destined to following IPs based on configured ACL rules
 - UE IP addresses
 - NAT IP addresses
 - Loopback interface addresses
 - Interface addresses
- The following are the supported ACL rule constructs
 - Permit/Deny
 - L3 - IPv4/IPv6 source and destination addresses
 - L3- IP proto
 - L4 - Source / Destination port
- For UPLINK packets (in N3 network context), ACLs are applied solely at the outer IP level and not on the inner subscriber UE packets
- For DOWNLINK packet (in N6 network context), ACLs are applied on subscriber packets (assuming they are not tunneled)
- Sample Configuration:

```
upf service-construct network-context N3
    security-policy N3_SECURITY_POLICY
    !

upf service-construct security-policy N3_SECURITY_POLICY
    acl-rule-group ACL_RG
    !

upf service-construct acl-rule-group ACL-RG
    admin-state      enable
    acl-rule ACL-RULE-1
    acl-rule ACL-RULE-2
    !

upf service-construct acl-rule ACL-RULE-1
    admin-state enabled
    priority 115
```

```
        action      allow/deny
        packet-filter 105-USVA-01
        packet-filter 105-USVA-02
    !
```

ADO-1620869

Config checks for UE Pools, NAT Pools and Loopbacks

This feature validates UE Pool, NAT Pool and Loopback configuration sent to the UPF and returns back the results. The purpose of this feature is to eliminate misconfiguration and provides operators a way to check commits as to how many routes would be generated before the actual commit is performed. This is accomplished by issuing CLI command '**commit check**'. The 'commit check' command displays all the information about the objects, but the most important information is the total number of routes generated. A success is returned if all the validations for the configuration pass. If errors are found in the commit-check a failure is returned.

ADO-835463

Support of 1K VRFs and 3K DNNs

This feature is intended to provide functional support for configuring and accommodating 1k VRFs and 3k DNN. This feature enhancement also involves code optimization to handling config modification at backend. No new CLIs are introduced as part of this.

Platform and Infrastructure Features

ADO-1558434

Ability to provision individual or NESA mechid accounts for PaaS user interfaces Halo-e Integration in PaaS

This feature is to integrate the UC PaaS UIs with Halo-E OIDC provider. Also to prevent security breaches and vulnerabilities in the UC PaaS environment, credentials must be removed from charts and deployments. **This feature is disabled by default.**

Feature Brief:

- Halo-e is an OpenID Connect (OIDC) compliant Single-Sign-On (SSO) solution. It will host all the corporate users.
- This feature is to enable Halo-e based authentication for the following PaaS GUIs:

New Features and Enhancements

- Grafana (deployed in fed-grafana)
 - Grafana (deployed in fed-kube-prom)
 - Thanos (deployed in fed-prometheus)
 - Prometheus (deployed in fed-kube-prom)
 - Alert Manager (deployed in fed-prometheus)
 - Alert Manager (deployed in fed-kube-prom)
 - Jaeger
 - Kibana
- It should be possible for corporate users to authenticate themselves and login (via Single Sign-On) into all the above-mentioned GUIs.

Feature Enablement:

By default, this feature is disabled in the applicable feds such as fed-paas-helpers, fed-kube-prom, fed-grafana, fed-prometheus, fed-jaeger, fed-kibana, fed-otel-collector, fed-elastic and fed-elastalert.

The feature is enabled by setting the flag `global.authType = "haloe"` in the `values.yaml` of the respective feds.

ADO-1558449

SHA256-4096 and Higher PKI Key Size

The purpose of this feature is to support a higher key size for PKI along with existing PKI algorithm and current key size. This feature allows customers to configure and generate private keys with larger key size and use either RSA or ECDSA algorithm for certificates. The configuration for Certificate CRO can be passed to select different PKI Algorithm and key size. The Cert-manager generates certificates from the configured algorithm and key size and stores them in K8s secret.

Prior to this feature, only SHA256-2048 bit key size was supported. With this feature, the following combinations are supported:

- SHA256-RSA 2048
- SHA256-RSA 4096(Default value)
- SHA256-RSA 8192
- EC-256

Note: With UC 5.2.0, the certificates generated from cert-manager, the private key will be generated with default algorithm of RSA and 4096 bit key size. Before UC 5.2.0, private key is generated with default RSA algorithm and 2048 bit key size.

ADO-1078164

Miscellaneous Routing – Netconf support for ribmgr and bgpMgr CLI

This feature provides support of retrieval of operational data for the Miscellaneous Routing feature using Netconf protocol for Ribmgr and BgpMgr components. Please refer to UnityCloud User Guide/UnityCloud Overview for more details.

Note: Netconf-console binary is not packaged as part of cfgmgr. It should be installed on any VM which has connectivity to cfgmgr-svc load balancer IP. Netconf service runs on port 830. Ensure port 830 is not blocked. If for any reason port 830 cannot be allowed, port 22 can be used for running netconf queries.

ADO-734007

Decoupling OPM from SMF/UPF

This feature removes the dependency of the SMF and UPF clusters on the OPM's Docker registry under normal operations. This does not remove the dependency on OPM during NF deployment, upgrade or configuration activities. A new Docker registry federation (fed-infra-dreg) is installed on the SMF and UPF clusters to host images instead of pulling from the OPM.

ADO-1741621

Transition from Banzai cloud kafka to Strimzi kafka

As part of this feature, we are moving from the Banzai Cloud Kafka cluster to the Strimzi Kafka cluster. The fed-kafka and fed-zookeeper components are being replaced with fed-kafka-strimzi, which is deployed ahead of fed-smf, fed-upf and fed-amf-mme in the namespace fed-kafka-strimzi.

ADO-1040968

Support Show CLI pipelining for read-only users

This feature grants read-only users the ability to utilize pipe commands for processing the output of show commands, including 'more,' 'begin,' and 'include' etc. Read-only users should be able to efficiently filter and manage show command output using these pipe commands.

ADO- 733992

Configuration Manager: Support for Secrets management from Hashicorp Vault.

As part of 5.1.0, Certificate support for mTLS for UPF and SMF has been delivered. In 5.2.0, Configuration Manager secrets can also be stored and retrieved from the Hashicorp Vault instead of configuring as helm variable in plain text format. From this release onwards, Configuration Manager secrets can be configured either as helm variable or can be fetched from Hashicorp vault, when vault is enabled. By default, the secrets are read from values.yaml.

For more details about configuration manager support for Hashicorp Vault, please refer to “Configuration Manager support for Hashicorp Vault” section in Deployment Guide.

Note: Secrets for other application services can also be stored in Hashicorp Vault, when enabled.

Unsupported and Tech Preview Features

This section lists the unqualified or unsupported features/functionality in this release.

Unsupported Features

Note: AKS cluster creation from the OPM GUI is not supported.

Platform and Infrastructure

- **REQS-12780:** Disk sizing for rook-ceph

AMF

- **REQS-12897:** Public Warning System (PWS)

Tech Preview Features

This section lists the Tech Preview feature(s) in the current release. These features will be tested and fully qualified only in a future release.

Tech Preview features provide early access to upcoming product features that are available in the release but are not functionally complete and are not currently fully supported. These features are available for preview in a lab or test environment only and not meant for production deployment.

Table 6. Tech Preview Features

Feature / Enhancement	Description
N/A	N/A

Restrictions and Special Considerations

The following restrictions and special considerations apply to this release.

Table 7. Restrictions and Special Considerations

Restrictions and Special Considerations	Description
Single cluster deployment	This release supports a single cluster containing Platform as a Service (PaaS) and 3GPP Cloud-Native Network Functions (CNFs).
Developing Solution's DsTest product	The version of Developing Solution's DsTest product used with validation is devsol-dsTest_5.5.20201005_amd64.deb.
Deploying Helm charts	When you deploy Helm charts via OPM or OPM Helm APIs, the OPM creates the directory structure for the federation on the Network File System (NFS) that is used for configuration-related information. If raw Helm commands are used instead of the OPM, you must create these directories before deploying the federation. For example, to deploy the AMF as fed-amf, the Operator must issue the following commands before the deployment: 1. Log in to the OPM via SSH. <pre># ssh root@<opm ip></pre> 2. Issue the following command: <pre># mkdir -p /filestore/fed-amf/cfg ; mkdir -p /filestore/fed-amf/web</pre> 3. Type exit to log out from the OPM.
Devsol	If you use Devsol, make sure you are on the latest version and that the scripts are updated (3GPP TS 23.501 Version 15.2.0. to 15.4.0) to match UnityCloud compliance.
The docker web image has been removed from the private docker registry.	The docker web image has been removed from the private docker registry, which was previously accessible at <opm>:8080. You may use CLI commands to view a list of images in the private docker registry: <pre># ssh root@<opm ip> # docker image ls # docker pull <opm ip>:5005/myimages:TestImage</pre>
OPM import and export of multi-master clusters is not supported	Please take release note that multi-master is not supported for csv export and import. Need to fill in multi-master from OPM Cluster configuration.
cna-istio_config" image location	For AKS deployments, the image path needs to be updated in deployment: <azure container registry>/cna-istio_config:2.2.0-46
Two hosts with the same IP and different ports are currently not supported	Istio configuration using IP addresses requires to use of unique IP addresses for routing of the message from the application to its istio-proxy sidecar and then to egress. This is more so as istio cannot open the different ports specified by the application dynamically.

Restrictions and Special Considerations	Description
In-service rollback for rook-ceph	Upgrade from release 4.3.0 to 5.1.0, upgrades ceph from v17.2.6 to v18.2.1 which is a major version upgrade. Rollback for rook-ceph from 5.1.0 to previous releases is not recommended.
Chf-Agent	Ignore chf-server-profile port and schema configuration parameters.
TDEF-Proxy DPI Interworking	application-group (along with the subset of applications) and flow-attribute are supported in the protocol-application-rule-group config. Note: Application-group Encrypted and Generic is not supported.
ISSU (In-service SW upgrade) for SMF	Graceful shutdown is not supported on most of the micro services of SMF. Only smfcc micro service supports graceful shutdown. In flight transactions for call processing may fail depending on which microservices are being upgraded during the transaction during ISSU. This may result in session setup failures during ISSU and/or session loss with session modification failures during ISSU. N to N+2/N-2 S/W Upgrade/Rollback for SMF is not supported N+2 SW release interoperability between SMF and UPF UC releases is not supported. K8 Skip level Version upgrades not supported for ISSU. For observability fed - fed-prometheus ISSU, if storage is set to local(disk attached to local), historical Prometheus data will be lost. Persistent storage should be configured to retain the data across ISSU for Observability feds – fed-prometheus, fed-elastic & fed-kafka. During ISSU, worker nodes CPU reservation (not utilization) may increase up to 98% due to pods being created and terminated, this should be only transient during ISSU and nodes CPU reservation should come back to normal.
Pod status	Pods will come up and stay in n-1/n state till all its dependencies, including mandatory OAM configuration, are met. Once the dependencies are met, pod will go to n/n state.

Upgrade Notes for UC5.2.0

Canary upgrade is not supported for: AMF, cMME, NRF or NSSF:

- After upgrade, the configuration changes for a 'new' entry in config schema should be reverted, before a rollback is attempted.
 - a. For example, suppose a 'new' value is added to an existing enum range in the config schema of the upgraded release.
 - b. After upgrade, this 'new' enum value is configured in config manager.

- c. If a rollback is triggered without reverting this 'new' enum value, then rollback will fail.

ADO-208650 - Performance suite failure for increase in IPPE pod CPU usage

Higher CPU utilization has been observed for VPP worker threads on platform RHEL compared to platform Mariner. Specifically, there is a 14% relative increase in CPU usage on RHEL, where 2M sessions are handling ~48K SM context updates while maintaining a throughput of around 120G. The default disabling of Transparent Huge Pages (THP) on RHEL is currently being considered the primary reason for the increased CPU usage, in contrast to platform Mariner where THP is enabled by default. Plans are in place to fix this behavior in upcoming patches.

Reference ADO-2080697 - vpp worker and IO thread avg utilization is ~14% more compared to Mariner vs RHEL VMs

ADO-2070202 – Alerta and related charts not used/required

Some feds are not used/required by all customers. For example, fed-elastalert is an optional deployment for some customers and in scenarios like this, users should uninstall or remove fed-elastalert. This can be done manually, or through the OPM GUI.

ADO-1980410 - SMF 5.1 Deployment Issue: Some fed-smf Pods Not Fully Ready After SMF 5.2 Deletion

This is a known issue; in the 5.2 release the confd version is 8.0.14. As part of this release, rollback namespace is removed. When users install 5.2 and then install 5.1, confd warns that rollback namespace is not present. Users must clean up the filestore and then install 5.1.

The issue won't be seen if 5.1 is already installed once on the cluster. Only in cases of a fresh cluster where confd with 8.0.14 is installed (5.2/main_int) and then installed with to confd 8.0 (5.1) is this noticed.

ADO-2052828 – Rollback not completed

The OPM didn't rollback all feds to previous version; during a NF ISSU SMF upgrade, a failure was observed due to 'wrong image' and rollback was triggered.

Skip rollback of Elastic

- Rollback is not supported for fed-elastic, fed-kibana and fed-kubedb_operator. Upstream Elastic does not support rollback of upgraded elastic. (It is day one behavior).

Restrictions and Special Considerations

- If users want to go to previous release of Elastic then all the feds (fed-elastic, fed-kibana, fed-kubedb_operator) need to re-deployed with the desired compatible versions.

Config Changes

Caution: The following section contains detailed information regarding the config changes seen in this release per feature. This section is not meant to replace the Deployment guide procedures but to inform customers of the config changes to consider before an upgrade/deployment of this release.

Legend:

- Mandatory parameters are in **bold** type
- Optional parameters are in *italics*
- Conditional parameters are in ***bold-italics*** type

SMF Features

ADO-1031811

SMF reject PDU session upon UPF failures/UE pool exhaustion

- CLI Path: smf service service-profile

Attribute	Type	Range/Length/Pattern	Default	Description
<i>session-reject-for-ue-ip-pool-exhaust-detection</i>	Enum	disabled enabled	disabled	This field controls enable/disable of session setup reject when ue ip pool is exhausted for a dnn
<i>http-error-code-for-ue-ip-pool-exhaust</i>	Uint32	400..504	504	HTTP Error code to use for session reject when ue ip pool is exhausted for a dnn
<i>sm-5g-cause-for-ue-ip-pool-exhaust</i>	Uint32	26..111	26	5G SM NAS cause code to be sent in session reject when ue ip pool is exhausted for a dnn

Config Changes

Attribute	Type	Range/Length/Pattern	Default	Description
<i>sm-4g-cause-for-ue-ip-pool-exhaust</i>	Uint32	64..129	73	4G cause code to be sent in session reject when ue ip pool is exhausted for a dnn

■ CLI Path: smfmonitor service

Attribute	Type	Range/Length/Pattern	Default	Description
<i>nfprofile-update-on-pfcp-connection-flip</i>	Enum	disabled enabled	disabled	This field controls NFProfile Update when last/first pfcp association status flips
<i>nfprofile-update-on-dnn-ip-pool-exhaust</i>	Enum	disabled enabled	disabled	This field controls NFProfile Update to exclude/include a dnn when ue ip pool exhaust is detected/cleared for the dnn

■ UPF Configuration: CLI Path: pfcp

Attribute	Type	Range/Length/Pattern	Default	Description
<i>pfcp-dnn-report-update-period</i>	Enum	0..3600	0	This field controls periodic report of DNN Usage Reports from UPF to SMF. Value of 0 indicates no updates (feature disable). Default value to enable is 30 seconds
<i>pfcp-dnn-report-threshold</i>	Enum	0..100	85	This field controls which dnn can be included in the dnn usage report from UPF to SMF. Any dnn with ue ip pool utilization >= this setting will be included in dnn Usage Report until its

Attribute	Type	Range/Length/Pattern	Default	Description
				utilization falls below the setting
<i>pfcp-dnn-node-report-max-size</i>	Enum	21..32768	32768	This field specifies the max size of dnn usage report that can be sent in a node report request message from UPF.

ADO-1031813

Support single DNN with NI only

The feature can be enabled or disabled by using the configuration parameters below. Changes to the following configurations will only impact new sessions and not ongoing ones.

- CLI Path: smf service

Attribute	Type	Range/Length/Pattern	Default	Description
<i>dnn-matching-type</i>	Enum	exact-match ni-only ni-with-malformed-oi	exact-match	This field will determine the DNN matching algorithm. The option allow, the whole DNN matching, matching only

ADO-1031817

Virtual DNN based on vrf and charging characteristics

The feature can be enabled or disabled by using the configuration parameters below. Changes to the following configurations will only impact new sessions and not ongoing ones.

- CLI Path: smf service-construct subscriber-mgmt-profile <profile name>

Attribute	Type	Range/Length/Pattern	Default	Description
<i>mapped-dnn</i>	String	Preconfigured DNN name.		The incoming DNN gets mapped to the configured DNN name, provided the

Config Changes

Attribute	Type	Range/Length/Pattern	Default	Description
				session uses this subscriber-mgmt-profile.

- CLI Path (New object): smf service-construct reporting-profile <reporting profile name> <policy | charging | upf | pcsf | udm | radius> dnn

Attribute	Type	Range/Length/Pattern	Default	Description
<i>discovery</i>	Enum	Incoming matched current	current	This field controls what DNN gets reported on given interface. Incoming – DNN served in the session create message. Matched – DNN that matched with incoming DNN Current – latest DNN after any mapping
<i>messaging</i>	Enum	Incoming matched current	current	This field controls what DNN gets reported on given interface. Incoming – DNN served in the session create message. Matched – DNN that matched with incoming DNN Current – latest DNN after any mapping

- CLI Path: smf service-construct subscriber-management-profile <management profile name>

Attribute	Type	Range/Length/Pattern	Default	Description
<i>reporting-profile</i>	String	Preconfigured reporting profile name.		This configuration selects the reporting profile

Attribute	Type	Range/Length/Pattern	Default	Description
				that need to used for session selecting give subscriber-mgmt-profile.

ADO-1031835

Support SEPP discovery from NRF and SEPP redundancy

- CLI Path: *smf service-construct sepp-response-code-template*

Attribute	Type	Range/Length/Pattern	Default	Description
sepp-response-code	Uint16	N/A	None	This field defines the error code returned by and SEPP upon which when received SMF shall try the alternate SEPP

- CLI Path: *smf service service-profile peer-nf-profile*

Attribute	Type	Range/Length/Pattern	Default	Description
peer-nf-profile	String	<i>sepproxy</i>	None	The value "sepproxy" is added to this field for SEPP peer profile
<i>sepp-response-code-template</i>	String	Name of sepp-response-code-template configured under service-construct	None	The field contains the name of the sepp-response-code-template

ADO-1031820

DN-AAA Radius UE IP assignment

- CLI Path: *smf service-construct subscriber-management-profile*

Attribute	Type	Range/Length/Pattern	Default	Description
<i>ue-ip-allocation-mode</i>	Enum	local radius	local	This field will determine how the ue

Config Changes

Attribute	Type	Range/Length/Pattern	Default	Description
				ip allocation mode should be.

As a part of the CLI enhancement, following CLI Path has been updated. The existing values of *accounting-per-session* have been automatically converted to *accounting-type*, with no functional impact.

- CLI Path: smf service-construct radius-accounting-profile

Attribute	Type	Range/Length/Pattern	Default	Description
<i>accounting-type</i>	Enum	qos-flow-based session-based	qos-flow-based	This field contains how the accounting type should be.
<i>accounting-per-session</i> [Deprecated]	Boolean	False True	False	Deprecated This field contains how the accounting type should be. True: session based. False: qosflow based. Now this same behavior is achieved using <i>accounting-type</i> attribute

ADO-1540188 LI Security

LI key cannot be helm variable with clear text

The feature can be enabled or disabled by using the helm configuration parameters as mentioned in the LI Deployment Guide.

UPF Features

ADO-620114

UE Hairpin Support

Performing an ISSU (In-Service Software Upgrade) from a version that does not support this feature to one that does, will not function reliably until all PPEs are updated to the newer release. This inconsistency arises because the previous version does not utilize E/W (GRE) for hairpin packets, whereas the new version relies exclusively on E/W (GRE) for handling these packets.

- CLI Path: `upf service-construct dnn <dnn-name>`

Attribute	Type	Range/Length/Pattern	Default	Description
<i>allow-ue-hairpin</i>	Boolean	True, False	False	Allow UE to UE communication

- CLI Path: `upf service-construct ppe-session-management-profile`

Attribute	Type	Range/Length/Pattern	Default	Description
<i>sm-bundle-allrb-msg</i>	Enum	enabled, disabled	enabled	Enable/Disable bundling of resourceblock messages from session manager to VPP to support UE2UE hairpinning. Default is enabled

ADO-1031858

PFD Syntax backward compatibility

As part of this feature no new CLI configuration or helm variable has been added, nor has any existing CLI configuration been changed. Therefore, this feature will not impact an ISSU to newer version or rollback to older version.

No changes have been made to the existing YANG model or YAML.

ADO-1063858

Support for SIP ALG

In case of ISSU (In-Service Software Upgrade) from a version that does not support this feature to one that does, will not have this functionality until PPEs are updated to the newer release. Once PPEs are on latest release, new connections for SIP ALG control sessions can be established.

In case of Software downgrade to unsupported version, existing connections for SIP ALG control sessions will be reset.

- CLI Path: `upf service-construct firewall-service-policy alg {tftp|ftp|rtsp|pptp|ipsecnattassist|sip}`

Attribute	Type	Range/Length/Pattern	Default	Description
<i>Sip</i>	Enum			A new enum value for 'sip' type has been introduced in list of supported ALGs

ADO-1329627

Change chunk sizing logic (large UE pools) when configured as zero

As part of this feature no new CLI configuration or helm variable has been added, nor has any existing CLI configuration been changed. Therefore, this feature will not impact an ISSU to newer version or rollback to older version.

No changes have been made to the existing YANG model or YAML.

ADO-1550326

CGNAT security – Phase 1

As part of this feature new CLI configuration has been added and no existing cli's have been modified.

Performing an ISSU (In-Service Software Upgrade) from a version that does not support this feature to one that does, will not have any impact on the overall PPE functionality.

- CLI Path: `upf service-construct acl-rule <rule-name>`

Attribute	Type	Range/Length/Pattern	Default	Description
Priority	Enum	1... 4096		Assign priority to acl-rule

Config Changes

Attribute	Type	Range/Length/Pattern	Default	Description
<i>Action</i>	Enum	Allow, deny		Allow/deny the traffic matching to acl-rule
<i>Packet-filter</i>	Leafref to Packet-filter			List of Packet Filters associated with this ACL rule

- CLI Path: `upf service-construct acl-rule-group <rule-group-name>`

Attribute	Type	Range/Length/Pattern	Default	Description
<i>aclRule</i>	Leafref to acl-rule			List of ACL rules for this rule group

- CLI Path: `upf service-construct security-policy <policy-name>`

Attribute	Type	Range/Length/Pattern	Default	Description
<i>acl-rule-group</i>	Leafref to acl-rule-group			Name of an acl-rule-group to attach

- CLI Path: `upf service-construct network-context`

Attribute	Type	Range/Length/Pattern	Default	Description
<i>security-policy</i>	Leafref to security-policy			Name of security-policy to attach to this network-context

- CLI Path: `upf service-construct flow-idle-timer-profile <profile-name>`

Attribute	Type	Range/Length/Pattern	Default	Description
<i>default-tcp-flow-fast-age-timeout</i>	Uint32	10 ... 3600		Configure tcp flow fast age timeout in seconds
<i>default-udp-flow-fast-age-timeout</i>	Unit32	10 ... 3600		Configure udp flow fast age timeout in seconds

- CLI Path: `upf service-construct flow-idle-timer-profile <profile-name> flow-idle-timer-rule <rule-name>`

Config Changes

Attribute	Type	Range/Length/Pattern	Default	Description
<i>tcp-flow-fast-age-timeout</i>	Uint32	10 ... 3600		Configure tcp flow fast age timeout in seconds
<i>udp-flow-fast-age-timeout</i>	Unit32	10 ... 3600		Configure udp flow fast age timeout in seconds

- CLI Path: `upf service-construct flow-rate-limit-profile <profile-name>`

Attribute	Type	Range/Length/Pattern	Default	Description
<i>new-flow-creation-rate-limit-ul</i>	Unit32	0 ... 2000000000	0	The flow creation rate on the uplink direction, indicates connections per second
<i>new-flow-creation-rate-limit-dl</i>	Unit32	0 ... 2000000000	0	The flow creation rate on the downlink direction, indicates connections per second

ADO-1620869

Config checks for UE Pools, NAT Pools and Loopbacks

As part of this feature no new CLIs have been added. Hence performing ISSU(In-Service Software Upgrade) from a version that does not support this feature to one that does, will not have any impact.

A new option '**check**' has been introduced in '**commit**' command to ensure that IP pools and loopback configuration is validated before the actual commit is performed at backend. Command returns 'success' if validation passes otherwise 'errors' are returned.

Sample Config –

```
admin@pod-cfgmgr-bbc7d4956-njrr8 04:56:01(config)# show full-configuration upf service-construct network-context N6
upf service-construct network-context N6
tcp-mss-adjust-ipv4          0
tcp-mss-adjust-ipv6          0
type                         N6
header-sanity-check          false
ignore-notify-bgp-peer-state-change false
nat-pool natPool1
nat-mode                     Many_to_One
```

Config Changes

```
nat-allocation-policy statistical
nat-filter address_port_dependent
max-ports-per-session 0
l3am-nat-ips-per-ppe 2
max-port-alloc-retries 0
!
nat-pool natPool2
nat-mode One_to_One
nat-allocation-policy statistical
nat-filter address_port_dependent
l3am-nat-ips-per-ppe 1
!
ip-pool n6Pool1
admin-state enabled
type UE_IP_Pool
range-start 11.1.0.0
range-end 11.2.0.0
route-summary-prefix-length 8
advertise true
advertise-pool-aggregate false
route-tag-aggregate 0
route-tag 0
l3am-chunk-size 32768
!
ip-pool n6Pool5
type NAT_IP_Pool
range-start 131.1.1.100
range-end 131.1.1.150
route-summary-prefix-length 0
advertise true
advertise-pool-aggregate false
route-tag-aggregate 0
route-tag 0
nat-pool natPool1
!
!
admin@pod-cfgmgr-bbc7d4956-njrr8 04:56:16(config)# commit check
Validation complete
admin@pod-cfgmgr-bbc7d4956-njrr8 04:56:22(config)#
Message from INFO at 2024-11-25 04:56:22...

{
  "Ip-Pools": [
    {
      "ChunkInfo": [
        {
          "Chunk-Size": 1,
          "Number-Of-Chunks": 1,
          "Number-Of-Primary-Routes": 1
        },
        {

```

```
        "Chunk-Size": 32768,
        "Number-Of-Chunks": 2,
        "Number-Of-Primary-Routes": 1
      }
    ],
    "Errors": "",
    "IsPoolValid": true,
    "Number-Of-Summary-Routes-Per-Ppe": 0,
    "Pool-Name": "n6Pool11",
    "Total-Number-Of-Routes-Per-Upf-Per-Peer": 6
  },
  {
    "ChunkInfo": [
      {
        "Chunk-Size": 2,
        "Number-Of-Chunks": 25,
        "Number-Of-Primary-Routes": 1
      }
    ],
    "Errors": "",
    "IsPoolValid": true,
    "Number-Of-Summary-Routes-Per-Ppe": 0,
    "Pool-Name": "n6Pool15",
    "Total-Number-Of-Routes-Per-Upf-Per-Peer": 50
  }
],
"Total-Routes-Per-Upf-Per-Peer": 56,
"Validation-status": "Valid"
}
admin@pod-cfgmgr-bbc7d4956-njrr8 04:56:22(config)# commit
Commit complete.
admin@pod-cfgmgr-bbc7d4956-njrr8 04:56:41(config)#
```

ADO-835463

Support of 1K VRFs and 3K DNNs

As part of this feature no new CLIs have been added. Hence performing ISSU (In-Service Software Upgrade) from a version that does not support this feature to one that does, will not have any impact.

No changes have been made to the existing YANG model or YAML.

ADO-1944155

UPF - VPP crash during call model ramp up

DPA was sending fib add/del messages to the VPP in batches of 1000 routes every 100ms, causing a mismatch in message pacing. VPP became unresponsive to the DPA client, leading to a

Config Changes

termination after 30 seconds of missed watchdog messages. CLIs are added to set the `fibSyncTime` and `fibSyncBatchSize` parameters. `fibSyncTime` is the duration in which DPA shall send fib add/del messages to VPP. `fibSyncBatchSize` is the no. of fib add/del messages can be batched to send to VPP with the interval of `fibSyncTime`. Based on the deployment, these parameters can be modified to manage the VPP Fib synchronization from DPA. DPA writes 50 fib add/del messages to VPP in every 100ms, by default.

- CLI Path: routing

Attribute	Type	Range/Length/Pattern	Default	Description
<i>fib-sync-time</i>	Unit32	100 ... 65535	100	VPP Fib Synchronous timer (ms). Recommend multiples of 100ms for every multiple of 50 of <code>fib-sync-batch-size</code> . Default: 100ms
<i>fib-sync-batch-size</i>	Unit32	100 ... 65535	50	Fib Batch size in an interval of (<code>fib-sync-time</code>) ms. Recommend multiples of 50 for every multiple of 100ms of <code>fib-sync-time</code> . Default: 50

PaaS Features

ADO-1270324

Post-recovery pod rebalancing

Note: This feature is not fully qualified and disabled by default. There is no need to change any values here.

Following helm configurations have been added to this feature:

Fed Name	Parameter Values	Description
fed-kube-addons	pod-descheduler: policy: removePodsViolatingTopologySpreadConstraintSmf: <i>enabled: false</i> includeSoftConstraints: true topologyBalanceNodeFit: true	A new profile has been added to the descheduler policy. This is an extension to the existing policy. This profile is responsible for descheduling smfcc pods in fed-smf

Config Changes

Fed Name	Parameter Values	Description
	labelSelector: matchLabels: app: smfcc-app namespacesIncluded: - "fed-smf" nodefit: true thresholdPriorityClassName: ""	namespace for violating topologySpreadConstraints. The profile is disabled by default. There is no need to change any other variables in this profile.
fed-smf	pod-smfcc: topologySpreadConstraints: <i>enabled: false</i> constraints: - <i>maxSkew: 1</i> <i>topologyKey: kubernetes.io/hostname</i> <i>whenUnsatisfiable: ScheduleAnyway</i> <i>nodeTaintsPolicy: Honor</i>	By default, this is disabled and reads the configuration from the global section. It will read and apply the global topologySpreadConstraints values for smfcc pods if this section is not present under pod-smfcc. To support descheduling of smfcc pods, the pod-smfcc now has its own config for topologySpreadConstraints. This is done to have a better control on the variables and tune it based on descheduler behaviour. Ex. At global level, the maxSkew maybe set 1, but for smfcc pods, this can be set to 2 to avoid descheduler pod eviction loop. Topology keys must be same as defined at global level. The only change will be to the maxSkew parameter. Also, note that the <i>nodeTaintsPolicy</i> is not present at the global level and must be added here for each constraint.

The following table describes each newly added helm variable.

Config Changes

Fed Name	Parameter Values	Default Value	Description
fed-kube-addons	pod-descheduler.removePodsViolatingTopologySpreadConstraintSmf	N/A	New descheduler profile added to monitor smfcc pods in fed-smf. smfcc pods use topologySpreadConstraints and only that plugin is enabled for this profile.
	pod-descheduler.policy.removePodsViolatingTopologySpreadConstraintSmf.enabled	false	Enabled(false) by default. Set to true to enable monitoring of smfcc pods for violation in topologySpreadConstraints.
	pod-descheduler.policy.removePodsViolatingTopologySpreadConstraintSmf.includeSoftConstraints	true	smfcc is configured with soft constraints (ScheduleAnyway) if the topology spread can't be guaranteed. This config instructs descheduler to look for soft constraint violations along with hard constraints.
	pod-descheduler.policy.removePodsViolatingTopologySpreadConstraintSmf.topologyBalanceNodeFit	true	This is needed for balancing the topology domains.
	pod-descheduler.policy.removePodsViolatingTopologySpreadConstraintSmf.thresholdPriorityClassName	Leave empty, no values to be set.	Only pods under the threshold can be evicted. smfcc pod has lowest priority. This can be left empty as only smfcc pods are monitored and need to be evicted if there is violation.
	pod-descheduler.policy.removePodsViolatingTopologySpreadConstraintSmf.labelSelector	N/A	Section for filtering pods within fed-smf.
	pod-descheduler.policy.removePodsViolatingTopologySpreadConstraintSmf.labelSelector.matchLabels	N/A	The object can contain multiple labels for selection.
	pod-descheduler.policy.removePodsViolatingTopologySpreadConstraintSmf.labelSelector.matchLabels.app	smfcc-app	This is set to smfcc. So only smfcc pods will be filtered in fed-smf.

Config Changes

Fed Name	Parameter Values	Default Value	Description
	pod-descheduler.policy.removePodsViolatingTopologySpreadConstraintSmf.namespacesIncluded	fed-smf	This is the list of namespaces that must be included. Only fed-smf is added to this list.
	pod-descheduler.policy.removePodsViolatingTopologySpreadConstraintSmf.nodefit	true	Set to true by default. The descheduler will consider whether or not the pods that meet eviction criteria will fit on other nodes before evicting them. If a pod cannot be rescheduled to another node, it will not be evicted
fed-smf	pod-smfcc.topologySpreadConstraints	N/A	This section is added to have an exclusive topologySpreadConstraints config for smfcc pods. Presently, this value is read from global section of values.yaml of fed-smf. If this section is present, the config will be read from here, else from global section. This needs to be configured for smfcc pod as the skew value can be different compared to the global value.
	pod-smfcc.topologySpreadConstraints.enabled	false	Set to true to enable this topologySpreadConstraints config on smfcc pods.
	pod-smfcc.topologySpreadConstraints.constraints	N/A	List of constraints for the pod.
	pod-smfcc.topologySpreadConstraints[*].maxSkew	1	Describes the degree to which pods may be unevenly distributed. You must specify this field, and the number must be greater than zero.
	pod-smfcc.topologySpreadConstraints.topologyKey	kubernetes.io/hostname	This is the key of node labels. Nodes that have a label with this key and identical values are considered to be in the same topology. We call each instance of a topology (in other words, a <key, value> pair) a domain. The scheduler will try to put a balanced number of pods into each domain.
	pod-smfcc.topologySpreadConstraints.whenUnsatisfiable	Schedule Anyway	Indicates how to deal with a pod if it doesn't satisfy the spread constraint. This should be set to ScheduleAnyway for smfcc pods.
	pod-smfcc.topologySpreadConstraints.nodeTaintsPolicy	Honor	Indicates how the scheduler will treat node taints when calculating pod topology spread skew. Set this Honor so that nodes without taints, along with tainted nodes for which the incoming pod has a toleration, are included. The descheduler also looks at this config for calculating the skew value.

ADO-1558434**Ability to provision individual or NESA mechid accounts for PaaS user interfaces****Halo-e Integration in PaaS**

Following helm configurations have been added to this feature:

Fed Name	Parameter Values	Description
fed-paas_helpers fed-grafana fed-kube_prom fed-elastic fed-kibana fed-elastalert fed-jaeger fed-otel_collector	global: authType: "local"	This field controls the authentication mechanism to be used for all PaaS UI components. By default, it is set to "local" and "adminPassword" field under global section must be set. It can be set to "haloe", if haloe authentication is required which requires Vault must be setup.

As part of the Haloe integration feature in PaaS Observability UI components (release 5.2 requirement), new HELM variable is introduced. The default is set to "local" to login to PaaS UI's, the current behaviour. If authType is set to "haloe", it requires Hashicorp Vault backend support.

The following feds now have an authentication type flag (authType):

- fed-paas-helpers
- fed-grafana
- fed-kube-prom
- fed-kibana
- fed-elastic
- fed-jaeger
- fed-otel-collector
- fed-elastalert

As part of OTEL Collector upgrade the jaeger protocol is no longer supported in OTEL Collector. Please use latest values.yaml config with OTLPHTTP config to deploy fed-otel_collector HELM chart.

Elastic Upgrade

Elastic and Kibana are upgraded as part of UC5.2.0 release. During the upgrade the following feds need to be upgraded and the respective values.yaml need to be updated:

- fed-kubedb_operator
- fed-elastic and fed-kibana - Config changes:
 - fed-elastic - elasticversion: xpack-8.14.1
 - fed-kibana - version: kibana-8.14.1

Fed Name	Parameter Values	Description
fed-elastic	global: <i>elasticVersion: xpack-8.14.1</i>	This field is to provide Elastic search version

Fed Name	Parameter Values	Description
fed-kibana	pod-kibana: upgrade: <i>version: "kibana-8.14.1"</i>	This field is to provide Kibana version

Infra Features

ADO-734007

Decouple OPM from SMF-UPF

A new federation, fed-infra_dreg starts a Docker registry that will host container images. The federation requires Cinder CSI storage on the cluster. In OPM, this can be enabled by setting "cloudprovider=openstack" and configuring its parameters. If using Heat to configure the cluster, all "cloudp_*" parameters must be set.

- values.yaml The following are some of the required parameters.

Config Changes

Fed Name	Parameter Values	Description
fed-infra_dreg	global: registry: docker: upstreamRepoPath: "changeme.opm.com:5005" infraDreg: enabled: true loadBalancerIP: "1.2.3.4" port: 5005 registryUrl: "changeme.com" secretName: "infradreg-cert-manager" storageSize: "1T" storageClass: "csi-sc-cinderplugin"	upstreamRepoPath: For fed-infra_dreg on SMF or UPF, this is the OPM docker registry FQDN. For MN-OPM, this is the FQDN to the ACR. Include port if needed. enabled: Enable the Docker registry feature loadBalancerIP: Assigned MetalLB IP to expose the registry service to the network port: Registry service port registryUrl: FQDN for the registry service that is resolvable from cluster's worker node secretName: Secret name generated by fed-paas-helpers cert manager. Default is infradreg-cert-manager storageSize: Size required to host all container images. Defaults to 1T storageClass: The Cinder CSI storage class. Defaults to "csi-sc-cinderplugin"

ADO-1558449 Support SHA256-4096 PKI Key Size

By default, cert-manger will generate certificate, with private key of 4096 bit size and encryption algorithm as RSA.

- values.yaml Changes for NF feds and feds where ingress and egress gateway used.

Fed Name	Parameter Values	Description
fed-smf fed-upf fed-grafana fed-nrf fed-prometheus ...	global: mesh: istio: ingressGwTLS: certManager <i>pkiAlgorithm: RSA</i> <i>pkiKeySize: "4096"</i>	pkiAlgorithm, pkiKeySize are newly introduced parameter. pkiAlgorithm: default "RSA" pkiKeySize: default "4096". Other accepted values are "2048", "8192" If pkiAlgorithm is set to ECDSA, pkiKeySize can only be set "256"

ADO-733995

Support Halo authentication for PaaS UIs

Fed Name	Parameter Values	Description
fed-paas-helpers fed-kube-prom fed-grafana fed-prometheus fed-jaeger fed-kibana fed-otel-collector fed-elastic fed-elastalert	global: authType: "local"	Enables Halo-E integration for these feds. This is enabled by default. To disable the feature, set <i>global.authType = "local"</i>
fed-paas-helpers	global: vault: enabled: true hashicorpVault: enabled: true serverUrl: <Hashicorp Vault Server URL> role: < Hashicorp Vault role to access secrets> authMountPath: <Auth mount path> vaultNamespace: <Hashicorp Vault namespace> secretNames: oidc: oidcClientIDSecretName: "" oidcClientSecretName: "" caCertSecretName: "" pod-oauth2_proxy: oauth2proxy: oidcIssuerUrl: "" oidcAuthorizeUrl: "" oidcUserInfoUrl: "" oidcTokenUrl: "" <i>cookieExpire:</i> <i>cookieRefresh</i>	Hashicorp Vault should be enabled and configured for fed-paas-helpers. This is a pre-requisite for this feature. The fields under "secretNames" are the key names in Hashicorp Vault that represents the client-name/client-secret (obtained after Halo-e registration) and the Halo-E's self-signed CA certificate (not required to be configured in vault if Halo-E certificates are signed by well-known CAs). Note: Whenever any of the above secrets in Hashicorp Vault are rotated, then the oauth2-proxy pod running in fed-paas-helpers needs to be restarted to pick-up the new value(s). The restart of this pod should not impact the call traffic. oidcIssuerUrl – Halo-E's URL oidcAuthorizeUrl: Halo-E's Authorize endpoint oidcUserInfoUrl: Halo-E's UserInfo endpoint oidcTokenUrl: Halo-E's Token endpoint cookieExpire - The time after which the 'cookie', that oauth2-proxy pod creates, will expire. This is set to '0' by default which means that the cookie will be a

Config Changes

Fed Name	Parameter Values	Description
		'session-cookie' which will expire when the browser is closed. cookieRefresh - OAuth2-proxy pod tries to refresh an Access-Token after the duration set in this parameter. Default value is 59m.

ADO-1741621

Transition from Banzai cloud kafka to Strimzi kafka

As part of this feature, we are moving from the Banzai Cloud Kafka cluster to the Strimzi Kafka cluster. The fed-kafka and fed-zookeeper components are being replaced with fed-kafka_strimzi, which is deployed ahead of fed-smf, fed-upf and fed-amf_mme in the namespace fed-kafka-strimzi.

Deployment steps:

Config Changes

Fed-name	Parameter	Value/Description
fed-opa	exemptedNamespaces	fed-kafka-strimzi (make sure its present)
Fed-kafka-strimzi	Release Name	Enter the release name.
	Namespaces	Enter the namespace for this federation.
	global	
	envVarsFedUniqueName	Enter a globally unique name.
	grafana > url	http://<<metalLB_PaaSIP>>:<port>
	pod-kafka_strimzi	
	broker > externalLbIp	<metalLB_PaaSIP1>
	broker > staticLbIps	<metalLB_PaaSIP2>
		<metalLB_PaaSIP3>
		<metalLB_PaaSIP4>
	Mirrormaker > enabled	<boolean>
fed-smf	pod-kafka_rplay_agt > kafkaClient > brokers	kafka-headless.fed-kafka-strimzi.svc.cluster.local:29092
	pod-chf_agent > kafkaClient > brokers	kafka-headless.fed-kafka-strimzi.svc.cluster.local:29092
	pod-smfcc > kafkaClient > brokers	kafka-headless.fed-kafka-strimzi.svc.cluster.local:29092
fed-upf	pod-ppe > kafkaClient > brokers	kafka-headless.fed-kafka-strimzi.svc.cluster.local:29092
	pod-proxy-> kafkaClient > brokers	kafka-headless.fed-kafka-strimzi.svc.cluster.local:29092
Fed-amf-mme	pod-amf_cc > kafkaclient > brokers	kafka-headless.fed-kafka-strimzi.svc.cluster.local:29092

Config Changes

Example Snippets:

1. By default, the fed-opa values.yaml excludes the fed-kafka-strimzi namespace from admission control. If not, update it to exclude this namespace.

```
pod-opa_gatekeeper:
  deployOpaGatekeeper: true
  podPriority:
    enabled: true
    priorityClassName: uc-priority-1
  exemptedNamespaces:
    - kube-system
    - fed-rbac
    - fed-rook-ceph
    - fed-kafka-strimzi
    - fed-zookeeper
    - fed-kafka
    - azure-arc
    - azurehybridnetwork
    - calico-apiserver
    - calico-system
    - kube-node-lease
    - kube-public
    - ssc
```

2. Update the values.yaml file for fed-kafka-strimzi with Grafana URL at global level.

```
grafana:
  enabled: true
  url: https://10.158.3.129:3000
```

3. Set the broker external bootstrap LB IP (externalLbIp) and broker IPs (staticLbIps) from the metallb subnet(s).

Config Changes

```
pod-kafka_strimzi:
  rbacEnabled: true
  serviceAccountNameAutoProxy: auto-proxy-sa
  serviceAccountNameKafkaOperator: kafka-operator-sa
  replicaCount: 1
  operator:
    leaderElection: false
    upgradeStrategy: Recreate
    resources:
      limits:
        cpu: 500m
        memory: 2Gi
      requests:
        cpu: 10m
        memory: 256Mi
  broker:
    externalLbIp: 10.158.3.149
    staticLbIps:
      - 10.158.3.150
      - 10.158.3.151
      - 10.158.3.152
```

4. After fed-kafka-strimzi is deployed, for new deployment or upgrading of fed-smf, update the kafkaclient in values.yaml as follows for pod-kafka_rplay_agt, pod-chf_agent, and pod-smfcc:

```
kafkaClient:
  brokers: kafka-headless.fed-kafka-strimzi.svc.cluster.local:29092
```

5. After fed-kafka-strimzi is deployed, for new deployment or upgrading of fed-upf, update the kafkaclient in values.yaml as follows for pod-ppe

```
kafkaClient:
  brokers: kafka-headless.fed-kafka-strimzi.svc.cluster.local:29092
```

6. After fed-kafka-strimzi is deployed, for new deployment or upgrading of fed-amf_mme, update the kafkaclient in values.yaml as follows for pod-amfcc

```
kafkaClient:
  brokers: kafka-headless.fed-kafka-strimzi.svc.cluster.local:29092
```

OPM Features

ADO-868232

OPM changes to remove use of Mongo DB creds storage as identified by crypto board

- No values.yaml changes in fed-opm and fed-opm_multi

ADO-734007

Decouple OPM & AOSM from SMF-UPF

- No values.yaml changes in fed-opm
- Values.Yaml Changes – fed-opm_multi

Fed Name	Parameter Values	Description
fed-opm_multi	<pre> global: .. registry: docker: repoPath: "<registry_docker_path>" infraDreg: enabled: true loadBalancerIP: "" port: 5005 registryUrl: "" secretName: "infradreg-cert-manager" storageSize: "1T" storageClass: "csi-sc-cinderplugin" </pre>	infraDreg – newly added attributes.

ADO-1534827

In-service rotation of Secrets with HashiCorp Vault- for both Network Cloud and Nexus (Non LI NFs and Modules)

- No values.yaml changes for fed-opm
- Values.Yaml Changes – fed-opm_multi

Config Changes

Fed Name	Parameter Values	Description
fed-opm_multi	<pre>global: .. vault: .. hashicorpVault: # Switch to enable / disable Hashicorp vault enabled: false # Hashicorp Vault server URL serverUrl: http://fed-hc-vault-test-svc-internal.fed-hc-vault-test:8200 # Hashicorp Vault role to access secrets role: "reader" # Hashicorp Vault Auth mount path authMountPath: "jwt" # Hashicorp Vault namespace that is optional vaultNamespace: "" .. secretNames: fedopmmulti: opmUsernameSecretName: secrets/data/opmUsernameSecretName opmPasswordSecretName: secrets/data/opmPasswordSecretName mongoUsernameSecretName: secrets/data/mongoUsernameSecretName mongoPasswordSecretName: secrets/data/mongoPasswordSecretName mongolnitUsernameSecretName: secrets/data/mongolnitUsernameSecretName mongolnitPasswordSecretName: secrets/data/mongolnitPasswordSecretName mongoConnectionStringSecretName: secrets/data/mongoConnectionStringSecretName</pre>	Vault and secretNames - newly added attributes for Hashicorp vault support.

Other Changes

There are some changes in fed-opm/fed-opm_multi values.yaml as part of the bug fixes,

- **ADO-2029986** OPM: ucreg advertised certs collide with infra-dreg cert on attached cluster when infraDreg set to true on mn-opm

Config Changes

values.yaml changes for fed-opm

Fed Name	Parameter Values	Description
fed-opm	global: .. registry: docker: repoPath: "<registry_docker_path>" opmDreg: enabled: true	Added a separate flag "opmDreg.enabled" to control the deployment of the OPM Docker registry pods.

values.yaml changes for fed-opm_multi

Fed Name	Parameter Values	Description
fed-opm_multi	global: .. registry: docker: repoPath: "<registry_docker_path>" opmDreg: enabled: true	Added a separate flag "opmDreg.enabled" to control the deployment of the OPM Docker registry pods.

Anomalies

This chapter describes the anomalies for this release. It is broken down into these sections:

- [New Anomalies in this release](#)
- [Existing Anomalies](#)

New Anomalies in this release

The following tables list the new anomalies per Network Function.

For information on features and enhancements included in previous releases, refer to the respective release notes.

SMF

The following SMF anomalies are added in this release.

Table 8. New Anomalies in SMF

Anomaly	Description
ADO-2002931 – Clone, IcM-584143332 Inbound Roaming UC vSMF to hSMF ims PDU Session Create request does not forward ePCO with pcscf ip request	In case of Inbound Roaming, during ims PDU Session Create request, vSMF does not forward ePCO IE with P-CSCF IP request towards hSMF. Workaround: N/A

OPM

The following OPM anomalies are added in this release.

Table 9. New Anomalies in OPM

Anomaly	Description
ADO-1975392 MN OPM: The cluster creation process is not updating the dreg cert configuration correctly.	The cluster creation process is not correctly updating the dreg certificate configuration when creating a cluster from MN OPM. Workaround: 1) The FQDN for the OPM dreg (infra-dreg FQDN if enabled) should be updated in the <code>/etc/hosts</code> file. 2) The dreg certificate should be copied to <code>/etc/containerd/certs.d/5005/</code> 3) The <code>/etc/containerd/config.toml</code> file should be updated with the dreg certificate location.
ADO-1998727 Fed helm options not set with top-level NF snapshot, causing ISSU upgrade request validation failure	The helmOptions were not set because only the top-level NF snapshot was provided, without including dependency feds. This incomplete snapshot caused the subsequent upgrade request validation to fail. In a normal scenario, the snapshot should include all feds, including dependency feds, to ensure the proper setting of helmOptions and successful validation of upgrade requests. Workaround: Manually update the helmOptions.
ADO-2002007 Rollback notifications should be excluded from the OPM GUI for fed-kubedb_operator and fed-otel-collector	During an ISSU failure, rollback notifications for fed-kubedb_operator and fed-otel-collector are displayed in the OPM GUI. Workaround: None; simply disregard the notifications.

Anomaly	Description
ADO-1942507 Kubernetes cert in the self k8rm pod is not renewed after the cert is updated	During a support activity to renew expired certifications it was observed that Cert-manager renewed the certs, however the self k8rm pod is not updated with the latest certificate. Once the self k8srm pod was restarted, the certs were updated. Workaround: Follow this MOP procedure: <u>kubeConfig (self kubernetes admin.conf cert) - Applicable for both fed-opm and fed-opm-multi</u> Fetches by k8rm/masterm during initialization and stored in /root/.kube/config. The m2mkey must be available to fetch this. Follow the “Kubernetes Certificate Renewal” section in <u>UC Deployment Guide</u> for the rotation. Once the Kubernetes certificate is renewed, follow the steps below to complete update for OPM. <u>Steps to Update the Renewed Self Certificate for OPM Pods:</u> <u>Restart the k8rm Pod (For OPM 4.3 to 5.1):</u> Restarting the self OPM k8rm pod updates /root/.kube/config with the renewed /etc/kubernetes/admin.conf. <u>Update Kubernetes Secrets (From OPM 5.2 Onwards):</u> Starting with OPM 5.2, the config is moved to Kubernetes secrets. Follow these steps to manually update the secret: -Identify the self kubeconfig secret: <pre># kubectl get secrets -n vnfmgr grep kubeconfig-def kubeconfig-def765-0123456789abcdef Opaque 1 47d</pre> -Encode the renewed admin.conf file: <pre># cat /etc/kubernetes/admin.conf base64 -w0</pre> -Update the kubeconfig secret: <pre>kubectl edit secret kubeconfig-def765-0123456789abcdef -n vnfmgr</pre> Replace the existing kubeconfig with the new base64-encoded data from the updated admin.conf
ADO-2050267 MN-OPM does not set up infra-dreg cert details on the attached NF clusters	Customers who do not use PADS should follow the MOP steps to set up the infra-dreg cert details before it can pull images from MN-OPM infra dreg.

Anomaly	Description
ADO-2050267 MN-OPM does not set up infra-dreg cert details on the attached NF clusters (continued)	Workaround: Follow these steps to configure OPM uc-reg (infra-dreg) in the NF cluster: 1) Retrieve infra-dreg loadBalancerIP and registryUrl: Run the following command to get the deployed values: <pre>helm get values -n fed-infra-dreg fed-infra-dreg grep -A 4 infraDreg</pre> Example output: <pre>infraDreg: enabled: true loadBalancerIP: 10.144.199.201 port: 5005 registryUrl: registry.unitycloud.com</pre> 2) Update /etc/hosts on all master and worker nodes. Add an entry for `<code><loadBalancerIP> <INFRA-DREG URL></code>` on each node: Example: <pre>echo "10.144.199.201 registry.unitycloud.com" >> /etc/hosts</pre>
	3) Copy certificates to all master and worker nodes. Copy `ca.crt` and `tls.crt` from OPM to all master and worker nodes. Ensure the target directories exist before copying. <pre>ssh root@<NF_node_IP> "mkdir -p /etc/pki/ca-trust/source/anchors"</pre> <pre>scp /etc/containerd/certs.d/<INFRA-DREG URL>/ca.crt root@<NF_node_IP>:/etc/pki/ca-trust/source/anchors/<INFRA-DREG URL>.ca.crt</pre> <pre>scp /etc/containerd/certs.d/<INFRA-DREG URL>/tls.crt root@<NF_node_IP>:/etc/pki/ca-trust/source/anchors/<INFRA-DREG URL>.tls.crt</pre> <pre>ssh root@<NF_node_IP> "update-ca-trust"</pre> <pre>ssh root@<NF_node_IP> "mkdir -p /etc/containerd/certs.d/<INFRA-DREG URL>"</pre> <pre>scp /etc/containerd/certs.d/<INFRA-DREG URL>/ca.crt root@<NF_node_IP>:/etc/containerd/certs.d/<INFRA-DREG URL>/</pre> <pre>scp /etc/containerd/certs.d/<INFRA-DREG URL>/tls.crt root@<NF_node_IP>:/etc/containerd/certs.d/<INFRA-DREG URL>/</pre> <pre>ssh root@<NF_node_IP> "systemctl restart containerd"</pre>

Anomaly	Description
ADO-2059178 MN OPM: cna_remove_inventory script fails to execute garbage collection for opm-dreg when both infra-dreg and opm-dreg are enabled	In 5.2 MN OPM, both opm-dreg and infra-dreg can be enabled, allowing images to be pushed to both the OPM local registry and the infra registry using cna_deploy_inventory. However, when images are removed using the cna_remove_inventory script, garbage collection is not executed for opm-dreg pods. Workaround: After running the `cna_remove_inventory` script, manually trigger garbage collection in the opm dreg pods, execute the following command for each vnfmgr-dreg pod: <pre>kc exec -it vnfmgr-dreg-0 -n vnfmgr -- sh registry garbage-collect /etc/docker/registry/config.yml</pre> Repeat the same for `vnfmgr-dreg-1` and `vnfmgr-dreg-2` pods.

Platform and Infrastructure

The following Platform and Infrastructure anomalies are added in this release.

Table 10. New Anomalies in Platform and Infrastructure

Anomaly	Description
ADO-2022209 fluentd-operator pod stuck in CreateContainerConfigError and deployment timed out for fed-paas-helpers	With vault enabled a config error is seen causing a stuck pod. The deployment then times out. Workaround: Add the following config in fed-paas-helpers values.yaml in the global section: <pre>secretNames: fluentd: mountCSI: true usernameSecretName: secrets/data/admin passwordSecretName: secrets/data/admin</pre>
ADO-2010251 pod-kargo stuck in container creating state when vault is enabled looking for admin-cred	When vault is enabled, there is no SPC for kargo which can pull secret from Vault. Workaround: Configure 'secret' through hashicorp/vault.

Anomaly	Description
ADO-2018438 fed-rook-ceph: helm status for fed-rook-ceph failed and ceph status in HEALTH_WARN state	Install failed at fed-rook-ceph. Ceph is in HEALTH_WARN state though all fed-rook-ceph pods are up. Workaround: If fed-rook-ceph install or upgrade fails because of failed Helm tests, users need to manually check the issue. To check, users need to perform the following steps. Note users must manually verify and fix Ceph issues to avoid more issues later on. 1. Check Ceph status <pre>kubectl -n fed-rook-ceph exec -it \$(kubectl -n fed-rook-ceph get pod -l "app=rook-ceph-tools" -o jsonpath='{.items[0].metadata.name}') -- bash ceph status</pre> a. If status shows HEALTH_WARN, due to a previous OSD crash, capture and archive crash logs. File a bug report with data from following commands. <pre>ceph crash ls ceph crash info <crash></pre> To mitigate issue: <code>ceph crash archive <crash></code> To verify issue is fixed: <code>ceph status</code> b. For any other HEALTH_WARN or HEALTH_ERROR status, file bug report with ceph status details if user cannot recover Ceph cluster. 2. Retry the install or the upgrade when Ceph is healthy.
ADO-2002749 Nexus-VO: SMF resiliency - Flexblock infra core on smfcc after a node reboot	In the case of node reboot scenarios, it is observed that Flexblock process inside the infra container restarts with an ELF core resulting in the pod restart where this issue happens. Workaround: None, there will be a pod restart.
ADO-2026575 Kargo approximate calculation is not even close to the completion time	On a rollup on an SMF cluster and as part of one of defect data collection started Kargo; it printed approximately 1M but it took 30mns to complete the job. Workaround: None.
ADO-2131781 After rollback, uninstall of fed-otel-collector failed due to clusterrolebindings.rbac.authorization.k8s.io "fed-otel-collector-crb" not found	After successful rollback of fed-otel-collector and fed-otel-operator, the old version of fed-otel-operator didn't have the predelete script. Hence it didn't try to patch opentelemetry crd. This lead to failure of fed-collector to delete the otel crd instance during its uninstall via bulk delete. Workaround: <code>root@sustspe-cna-mlkstuk-uc02-smf-master-01-1l18xq-qxd06ws29focuc3 [~]# kc -n fed-otel-collector delete opentelemetrycollector.opentelemetry.io "fed-otel-collector" deleted</code> <pre>root@sustspe-cna-mlkstuk-uc02-smf-master-01-1l18xq-qxd06ws29focuc3 [~]# kc delete ns fed-otel-collector namespace "fed-otel-collector" deleted</pre> Now continue with bulk delete operations

Anomalies

Anomaly	Description
ADO-2164573 VM-ISSU for fed-upf failed (worker node 3 didn't join back cluster after upgrade)	The VM ISSU for fed-upf failed after a worker node did not join back after the cluster upgrade. Workaround: None, after applying MOP to recover failed node VM ISSU resumed from that point.

UPF

The following UPF anomalies are added in this release.

Table 11. New Anomalies in UPF

Anomaly	Description
ADO-2010618 UPF PPE, investigate the ACL performance issues - increase in max CPU and avg CPU usage in case of ACL6. Also, need to check very high time taken to program large volume of rules	• When testing with ACL for IPv6 traffic, there is a CPU usage increase of about 10% for both the IO and worker threads of the PPE pod. • When we configure a huge number of ACL rules (like > 25 per network context), the time taken for all these rules to be programmed is very high causing a dip in the flow creation rate and traffic throughput Workaround: 1) Configure ACL only for IPv4 traffic 2) Configure few rules (like < 25 per network context) to ensure there is no delay in programming these rules

Anomaly	Description
ADO-2019250 failure of fed-upf upgrade due to "cannot patch "ppe-pdb" with kind PodDisruptionBudget"	During the Kubernetes upgrade on the Nexus UPF cluster, it was observed that up to three IPPEs could be unavailable at any given time. NAKS VMs upgrade quickly, taking only about 4 minutes to become ready and drain the old VM. However, when an IPPE moves to the new node, it takes around 10–11 minutes to be ready. This creates a situation where there can be up to three IPPE pods in the "n-1/n" state. Workaround: Edit the ppe pdb (PodDisruptionBudget) and set maxUnavailable: 1. <pre>kubect! get pdb ppe-pdb -n fed-upf -oyaml apiVersion: policy/v1 kind: PodDisruptionBudget metadata: annotations: meta.helm.sh/release-name: fed-upf meta.helm.sh/release-namespace: fed-upf creationTimestamp: "2025-01-24T07:19:51Z" generation: 2 labels: app: ppe-app app.kubernetes.io/managed-by: Helm name: ppe-pdb namespace: fed-upf resourceVersion: "3878047" uid: 8a2b5add-aaa2-400e-9a7c-c94020e42838 spec: maxUnavailable: 1 selector: matchLabels: app: ppe-app</pre>
ADO-2176716 Nexus: SMF ISSU - Drop in MBR rate by 50% during upgrade of fed-smf with LTE call model	During the ISSU on the SMF - when the ULB pods are going through the process of an upgrade, a more than expected rate of failure may be observed with the LTE MBR transactions - resulting in the sessions being subsequently closed. Workaround: None

Existing Anomalies

This section lists the anomalies from previous releases that still exist in this release.

AMF

The following are existing anomalies in the AMF.

Table 12. Existing Anomalies in AMF

Anomaly	Description
ADO-895660 SCTP failover not occurring as expected when all virtual Interfaces of a ulbvpp pod goes down	The SCTP connection does not survive intermittently when all virtual interfaces of a ULB are down. Workaround: <i>Ensure that the peer node SCTP profile is configured such that retries/timer-duration are longer than NHR route-failure detection but lower than N3IWF retries/timer duration.</i>

Common Microservices

The following are existing anomalies in Common Microservices.

Table 13. Existing Anomalies in Common Microservices

Anomaly	Description
ADO-466820, CN-42978 Show Diameter peer summary not showing any details when more than 100 peers are added to the system	Show Diameter peer summary not showing any details when more than 100 peers are added to the system. Workaround: <i>None</i>
ADO-466818, CN-43051 CN_MME Traffic Capacity: Show gtpc peer command not displaying any details when 100 peer SGWs are connected to the system	The CLI command does not support a huge amount of data to be displayed (For example: 100 peers) due to the limitations of the size of the message buffer provided by the platform for the application to respond back with required data to CLI. To support huge data to be displayed as part of this operational CLI, further enhancement to the Operational CLI functionality is needed and is planned for future release. Workaround: <i>None</i>
ADO-467164, CN-47830 The Alarm "Frequent Restart Unstable Network Function" is getting cleared without being set	The alarm log with the "Cleared" status is generated even if the alarm was not SET initially. Workaround: <i>None; this is just an event-log that can to be ignored (the status indicates "Cleared") as there is no corresponding SET operation. There is no functionality impact due to this event-log.</i>
ADO-436041, CN-46323 PFCP-U Handling of Accepting Association Setup Request from pfcpc-c in Going Down state.	The PFCP association does not establish once the PFCP-U pod is stuck in the GoingDown State. Workaround: <i>Restart the redis pods.</i>

Table 14. Existing Anomalies in Lawful Intercept (LI)

Anomaly	Description
ADO-433726, CN-31699 [SMF] Limitation with fed-istio cannot execute a certain PDSR based test case for the next IOT	When SMF and UPF are on the same cluster for the X2 delivery type DID, UPF failed to establish the connection due to istio-proxy interception. Workaround: <i>None</i>
ADO-505932, CN-31700 [UPF] Limitation with fed-istio cannot execute a certain PDSR based test case for the next IOT	When SMF and UPF are on same cluster for the X2 delivery type DID, UPF failed to establish the connection due to istio-proxy interception. Workaround: <i>None</i>

DPI

Table 15. Existing Anomalies in DPI

Anomaly	Description
ADO-906760 PPE should check operational DPI signature version as part of initialization	This issue is only seen after the first DPI signature package upgrade. The release cadence for Unity Cloud DPI Signature package artifact is once every two months. If a PPE pod restart is required at the UPF Network Function for any reason and if DPI signature package was already upgraded at the PPE pod, the restart operation would result in PPE coming up with the default DPI signature package version. If PPE pod is scaled up at the UPF Network Function, the new PPE replicas will come up with the default DPI signature package version. Workaround: <i>Trigger an upgrade to the impacted PPE using the operator CLI command “ppe v1 dpi upgrade dpi-image-version <dpi signature package artifact> podid <pod-id>” to restore the DPI signature version to the one that was running before the restart.</i>

N3IWF

Table 16. Existing Anomalies in N3IWF

Anomaly	Description
ADO-910448 N3IWF-IKEMGR pod restart seen during IPv6 - 1K UEs test run cna-ike_mgr-rel_4_3_0-sym:9 build	Sometimes IkeMgr pod's istio side car may crash with Out Of Memory cause, due to low memory limit set during the deployment.. Workaround: <i>It is recommended to set this value to 2 GiB during the deployment of fed-n3iwf namespace.</i>
ADO-1316993	IPSec ESP packets are dropped at Nwu interface while sending downlink IPSEC ESP Packet drop issue traffic towards UE. Due to this, N3IWF functionality will not work

Workaround: None.

NRF

The following are existing anomalies in NRF.

Table 17. Existing Anomalies in NRF

Anomaly	Description
ADO-433589, CN-34985 NRF returns 404 NF Profile Not Found for discovery with supported query parameters like ServiceName, even when NFs with matching serviceName under nfServiceList is Registered to NRF	The NRF does not look in Map-based IEs for lookup. <i>Workaround: None</i>

OPM

The following are existing anomalies in OPM.

Table 18. Existing Anomalies in OPM

Anomaly	Description
ADO-682376 Create cluster NBAPI does not support all configuration.	Create cluster NB API does not support arc_enable, hugepages_enable and disk_mode. <i>Workaround: None</i>
ADO-988636 Snapshot fetches the other charts in kubsystem namespace on NAKS cluster	On NAKS cluster, there are some system specific charts which are fetched while OPM fetches the charts for Snapshot creation. <i>Workaround: Workaround for this issue is while creation of Snapshot of NAKS cluster unselect all the charts from kubsystem in the PaaS Components box.</i>
ADO-1470458 fed-smf externalGtpcSvc data is not populated from shared parameters	Create a NF config set for fed-smf, externalGtpcSvc values cannot be populated from shared parameters. <i>Workaround: User must manually populate values for externalGtpcSvc in fed-smf chart, then save the config set.</i>
ADO-1993487 ISSU never completes and cannot abandon	Under some unknown circumstances, an ISSU can hang and never complete. This leaves the redis cache in a state where it cannot be abandoned. To verify if this has occurred, look at the redis-cache keys to see if there is an ISSU request that has never completed.

Anomaly	Description
	Workaround: OPM 5.2 supports end-to-end abandon. Use the Abandon option and select Yes in the confirmation popup to abandon an on-going ISSU operation.

Platform and Infrastructure

The following are existing anomalies in Platform and Infrastructure.

Table 19. Existing Anomalies in Platform and Infrastructure

Anomaly	Description
ADO-433703, CN-32184 fed-db-etcd etcd- cluster is not recovering after triggering a liveness issue	The fed-db-etcd etcd-cluster is not recovering after triggering a liveness issue. Workaround: If only one pod is in 1/2 (Not Ready) state, kill that particular pod and the etcd-operator will instantiate a new pod.
ADO-434092, CN-25440 fed-db-etcd pods do not support dead office recovery (all nodes simultaneous power cycle)	An etcd cluster automatically recovers from temporary failures (Example: machine reboots) and tolerates up to (N-1)/2 permanent failures for a cluster of N members. When a member permanently fails, whether due to hardware failure or disk corruption, it loses access to the cluster. If the cluster permanently loses more than (N-1)/2 members then it disastrously fails, irrevocably losing quorum. Once quorum is lost, the cluster cannot reach consensus and therefore cannot continue accepting updates. Workaround: Perform the following steps to recover from disastrous failure: If some etcd pods are up, first delete all etcd pods: <code>kubectl delete pod -n fed-db-etcd <pod-name></code> Edit the etcd-operator deployment and change spec.replicas to "0": <code>kubectl edit deployment -n fed-db-etcd fed-db-etcd-pod-etcd-operator-etcd-operator</code> Save the CRO etcdcluster yaml to <i>/tmp</i>, delete that CRO, and then redeploy it: <code>kubectl get etcdcluster -n fed-db-etcd -o yaml > /tmp/etcdcluster.yaml</code> <code>kubectl delete etcdcluster -n fed-db-etcd etcd-cluster</code> Edit <i>/tmp/etcdcluster</i> and remove the data under the subsection "status" <code>kubectl create -f /tmp/etcdcluster.yaml</code> Edit the etcd-operator deployment replica back to "1": <code>kubectl edit deployment -n fed-db-etcd fed-db-etcd-pod-etcd-operator-etcd-operator</code>

Anomalies

Anomaly	Description
ADO-436861, CN-31902 Container/pod outage of ~60 secs due to liveness failure	When an application gets into a hung state, with the current liveness probe configuration in place, there could be a maximum of two minutes service outage before the service is restored in the form of a new application pod. Workaround: None.
ADO-433716, CN-31916 No Liveness implementation: fed-istio citadel/pilot	This is an enhancement to support readiness/liveness in third party software. The need of readiness/liveness in istio containers and how to implement them is currently under evaluation to ensure forward compatibility. Readiness/liveness is to reduce the pod outage duration so that the subscriber outage is minimum. These functionalities are not mandatory for these PaaS pods. Workaround: None.
ADO-433715, CN-31921 No Liveness implementation for pods: fed-db-etcd and etcd-operator	This is an enhancement to support readiness/liveness in third party software. The need of readiness/liveness in istio containers and how to implement them is currently under evaluation to ensure forward compatibility. Readiness/liveness is to reduce the pod outage duration so that the subscriber outage is minimum. These functionalities are not mandatory for these PaaS pods. Workaround: None.
ADO-433701, CN-32188 No Liveness implementation for fed-kafka	This is an enhancement to support readiness/liveness in third party software. The need of readiness/liveness in istio containers and how to implement them is currently under evaluation to ensure forward compatibility. Readiness/liveness is to reduce the pod outage duration so that the subscriber outage is minimum. These functionalities are not mandatory for these PaaS pods. Workaround: None
ADO-433711, CN-32069 Liveness not implemented on fed-istio/istio-policy	This is an enhancement to support readiness/liveness in third party software. The need of readiness/liveness in istio containers and how to implement them is currently under evaluation to ensure forward compatibility. Readiness/liveness is to reduce the pod outage duration so that the subscriber outage is minimum. These functionalities are not mandatory for these PaaS pods. Workaround: None
ADO-433659, CN-33184 Lawful Intercept (X2 and X3): ipv4 is supported but not ipv6	For Lawful Intercept (X2 and X3) functionality, ipv6 is not supported. Workaround: Limit usage to ipv4 only.

Anomalies

Anomaly	Description
ADO-434264, CN-17287 Kibana index query and creation fails in elastic from Kibana after certain days	Observed a failure in elastic client pod after running fed-elastic for few weeks. After few weeks of running fed-elastic, the Kibana index query and creation from the elastic are failing. Workaround: <i>Restart the kibana pod.</i>
ADO-434072, CN-25945 Envoy in IPPE pod core dumped and restarted during traffic run	The envoy application in istio-proxy container of ippe pod experiences a segmentation fault under an undiagnosed but rare corner case leading to a core dump and restart of that container. End-to-end traffic is not observed to be impacted by it due to retry mechanisms in place for in-flight transactions. Workaround: <i>None.</i>
ADO-434004, CN-27562 UPF (fed-upf-2.2.0-264) IPPE's Istio-Proxy Envoy Core	Under current default Istio installation options, a yet undiagnosed corner case leads to a crash and coredump in istio-proxy (envoy) container. When this happens, Kubernetes restarts the istio-proxy container which is expected to resume its procuring role immediately. Workaround: <i>To remove the possibility of this specific type of problem in envoy, fed-istio can be launched by setting the value of <code>global.disablePolicyChecks</code> variable to true. Doing this does not affect any known istio functionalities used in UnityCloud.</i>
ADO-436709, CN-34512 istio-ingress gw 503 response format issue	If the SMF is reachable at its IngressGW load-balancer address, but this SMF has no reachable instances of the SMFCC application, the IngressGW returns a 503 plain/text error to the AMF. The AMF prefers to see an "application/json" response. This anomaly is observed under engineered negative test case testing. Workaround: <i>None.</i>
ADO-417875, CN-36683 Pods are not recovering after worker node failure test (K8s version v1.21.2)	Pods are not recovering after worker node failure test. Workaround: <i>None</i>
ADO-436567, CN-37190 Pods in a reboot loop after worker node failure test with superchart	This defect only applies to clusters with two worker nodes deployed. This defect does not apply to clusters with three or more worker nodes. With two worker nodes, there are deficiencies in node reboot testing. The fed-service-reg pods will not always recover when one of the two worker nodes is rebooted. This may result in other pods on this recovering worker node being unable to properly start. Workaround: <i>None.</i>
ADO-433279, CN-44189 smfcc kafka-client container terminates during call model when EDR is enabled	If EDR is enabled for SMF and the rate is high, a pod restart of the Kafka sidecar is observed. Workaround: <i>There is no workaround. However, disabling EDR is an option to avoid the pod restart.</i>
ADO-433303, CN-43165 5g-viz filter based on smf session id has some of the headers	Wireshark (tshark) is not able to decode http headers for some packets. This is a known limitation of Wireshark (tshark).

Anomaly	Description
filtered as <unknown> in the filtered PCAP	Workaround: No workaround available until next Wireshark (tshark) version is released.
ADO-580908, CN-48865 Not able issue "kubectl drain" commands on nodes with chronos timers	While issuing "kubectl drain" commands in Graceful/Planned shutdown of worker node case some commands have issues on chronos timer pods. Workaround: Modify the MOP for Planned / graceful shutdown during maintenance window. To be able to drain Chronos pods provide the force option.
ADO-580907, CN-48933 Chronos timer pods are not restarted back on amf, nrf & nssf on complete cluster shutdown for more than 30 min	The Chronos Operator awaits notifications from Kubernetes when existing pods change state (e.g. from "scheduled" to "running"), so it can begin the configuration process. However, when the pods enter the state "Shutdown" (such as when a Kubernetes worker node has been manually turned down), Kubernetes chooses not to start the pod, and so doesn't send any notification to the Chronos Operator, resulting in a stuck state. This has been seen after a whole Kubernetes worker node is shutdown, stopping all services from running entirely - so this issue does not directly stop live service. Workaround: Delete the timer pods to force a Kubernetes notification to be sent to the Chronos Operator.
ADO-920829 cfgmgr is getting stuck while deleting VRF config from upf	There is a delay of around 2.5 mins while deleting a single VRF from config and committing the change. For example, when executing the following commands, the commit would take around 2.5 mins: no upf service-construct network-context VRF_1 commit Workaround: None
ADO-929915 No virtual restarts triggered when fed-db-etcd loses and get back its quorum resulting failure in call restoration	When ETCD quorum is lost due to multiple worker nodes going down, the system could get stuck in a bad state resulting in service impact. Note there won't be any NF virtual restart that are triggered automatically. Workaround: Trigger the NF virtual restart manually.
ADO-1506978 Rebooting one of the zones on UPF cluster resulting in Virtual Restart for fed-upf and sessions are lost	During Zone reboot that affects k8s Master nodes, the k8s etcd pods may perform a leader election. During the period of time it take haproxy to fail over other nodes attempting to do heartbeat to kube-apiserver may temporarily be marked "not-ready". This behavior impacts the pods like those of fed-db-etcd with node-not-ready toleration set to 0. As a result, etcd pods will be evicted from those nodes, causing the etcd cluster to lose quorum and forcing a virtual restart. This is an intermittent issue. Workaround: None

SMF

The following are existing anomalies in SMF.

Table 20. Existing Anomalies in SMF

Anomaly	Description
ADO-448176, CN-25934 If smfmonitor pod restarted in REGISTERED state, there can be a partial update followed by a complete nfprofile update.	If all the critical services are up and LB IP is also available before the config data, then NFUpdate going with partial NFProfile is seen. During testing this issue did not occur, as config data always available before the LB IP and critical services. However, there is a possibility that it can occur again. Workaround: None
ADO-1340755 Timer pods were not part of post upgrade hooks during SMF Federation ISSU	Timer pods running in SMF Federation are upgraded serially at the end of SMF ISSU and are not considered part of ISSU cycle. Each timer pod takes around 5 minutes for upgrade which is dependent on the number of timers running. Revalidation and rejoin timers are external timers that run on timer pod. The number of timers running will increase the time taken to complete ISSU. Workaround: None
ADO-1402524: PcfUpdatePolicyRules3gppProcCtxt is not pended on collision with Bearer Resource Command	On running CbrBrccollision and UbrBrccollision in pgwcc-brc-suite.py, it was observed that the BRC is not processed immediately. Instead, the BearerResourceCommand procedure is pended. Workaround: None

UPF

The following are existing anomalies in UPF.

Table 21. Existing Anomalies in UPF

Anomaly	Description
ADO-434023, CN-27108 AUC 221-581 - UPF - ippe pods status change from 5/5 to 4/5	Due to conditions still under investigation, the memory consumed by the ppe or ippe pod in fed-upf may exceed its default limit causing it to be killed. Workaround: To help alleviate the chances of this happening, it is recommended to increase the memory limit by setting the helm variable pod- ppe.guaranteedScheduling.proxyMemoryLimit in fed-upf to 8Gi.

Anomalies

Anomaly	Description
ADO-434282, CN-16090 All fragmented packets are sent to the DPI module without reassembling	Fragments of a packet are processed by DPI as individual packets, resulting in erroneous DPI detection and statistics. Workaround: <i>UnityCloud 1.3.0-19 adds the basic support for Jumbo frames. There are a few limitations in what level of fragmentation and reassembly is supported in this release.</i> <i>Outer GTP-U tunnel fragmentation on N3 is limited to UPF clusters with single VPP worker nodes. Fragmentation and reassembly of the inner packets on the N3 tunnels are not supported (only works with unfragmented inner packets).</i> <i>Fragmented packets on N6 interfaces are supported.</i> <i>Full support for outer tunnel fragmentation on N3 will be supported in a future release.</i>
ADO-433692, CN-32412 Robustness: Worker node reboot causing long outage in upf	When a worker node is rebooted, session creations are failing for 2-3 minutes. Workaround: <i>No workaround available.</i>
ADO-436948, CN-30253 Routing and session programming out of sync leading to packet loss	On PPE pod failure, the sessions running on the PPE is restored after the routing, which results to a traffic dip until network convergence and restoration of DB on new PPE. Workaround: <i>No workaround available.</i>
ADO-465671, CN-31918 PODs no Liveness implementation - UPF - fed-upf - istio-egressgateway/istio-ingressgateway	Evaluating the need of readiness/liveness in these containers and how to implement them so forward compatibility is not affected. Readiness/liveness is mainly to reduce pod outage duration so subscriber outage is minimum. These are not application pods so these functionalities are not mandatory here. Workaround: <i>No workaround available.</i>
ADO-433540, CN-36121 – Clone VPP Missing Session Attributes After Restore - DNN and Tethering attributes	DNN and Tethering session attribute are missing after session restoration. Workaround: <i>None</i>
ADO-662347 VPN routes are not deleted from ribmgr after deleting the VRF configs	After a VRF delete config was performed, the VPN routes become inactive in rib (not deleted from rib table). Workaround: <i>None; no functional impact.</i>
ADO-1320726 Issue while retrieving frame queue entry with thread index 65535	Sometimes during handover of IPV6 fragments in load scenario the thread index (to which packet is to be handed over) is invalid. Workaround: <i>None</i>
ADO-1497514 VPP issue while removing routing & IM config with VRFs	VPP coredump is seen while removing and readding routing config Workaround: <i>None</i>

Anomalies

Anomaly	Description
ADO-731860 Routing: Stale entries are present in FTN table with old exp bit	Stale entries are present in FTN table with old exp bit when dynamically the MPLS exp bit is changed Workaround: <i>Dynamically changing MPLS exp bit is not recommended, it must be deleted and added in two different commits.</i>

Anomalies Fixed

This chapter lists the anomalies fixed in the main GA release and the patch stream.

Anomalies fixed in UC5.2.0

OPM

The following OPM anomalies are corrected in this release.

Table 22. Anomalies Corrected in OPM

Anomaly	Fix Description
ADO-1920491 OPM upgrade to 5.1.0.10 failed due to large number of evicted pods	Introduced continuous monitoring and deletion of evicted pods every 60 seconds to address issues with large numbers of evicted pods during OPM upgrades.
ADO-1857018, lCM-549423359 MN-OPM reboot support	Added priority settings to MongoDB pod members in the replica set initiation command.
ADO-707244 Unable to run cna_deploy_inventory on the multi-master, deployed on cluster created by OPM.	- Added copy_m2mkey_to_cluster_nodes function in working/scripts/masterServer.py to copy m2m keys to cluster nodes. - Introduced a new REST API endpoint /vnfmgr/api/v1/cluster/<cluster_id>/copyM2MKey to trigger the m2m key copying process.
ADO-1475369 Not able to load sharedValues in a Snapshot	Added code to ensure the correct class is checked before accessing treeNode data.
ADO-2044864 VM-ISSU: Some pods in fed-infra-dreg struck in Pending/Init:ImagePullBackOf state after VM-ISSU on SMF cluster.	Corrected etcHostsCtl.sh script to correct regex wildcard issue overwriting mnopm FQDN.
ADO-2048073 OPM_USER and OPM_PASSWORD env variables required in k8rm pod for OPM secrets rotation	Introduces environment variables for OPM username and password in the k8rm pod to enable dynamic rotation of OPM secrets from HashiCorp Vault.
ADO-2040627 NF- ISSU got stuck and not proceeded to SMF federation	Added retry logic and enhanced logging. Also added checks and logging for null deployment zone objects.

Platform and Infrastructure

The following Platform and Infrastructure anomalies are corrected in this release.

Table 23. Anomalies Corrected in Platform and Infrastructure

Anomaly	Fix Description
ADO-2020633 fed-infra-dreg: /etc/hosts in some of masters are not updated with "registryUrl" configured, leading "cert-updater" pods struck in Init state	Fixed in latest rollup; not seen in latest rollup.
ADO-2022932 fed-elastic pod failed to attach volume	Application pods that mount an Openstack Cinder volume may fail to come up with a FailedMount warning and error log that says "mkfs.ext4: Device size reported to be zero. Invalid partition specified, or partition table wasn't reread after running fdisk, due to a modified partition being busy and in use. You may need to reboot to re-read your partition table." This occurs because of a race condition between kubelet and Openstack Cinder. To mitigate the issue, reboot the node that is hosting the affected pod.
ADO-2031877 Kibana pod crash seen while generating csv report for searched logs (around 500 row logs)	In kibana, csv export is not supported; collection of logs should be done through kargo.
ADO-2054054 - Clone ppe-session-events topic is missing in default values.yaml file under fed-kafka_strimzi chart	Added ppe-session-events topic in interface/startfedvalues.yaml and interface/values.yaml. Adjusted partition and replication factors for multiple topics in interface/startfedvalues.yaml. Increased memory limit for pod-kafka_strimzi in interface/values.yaml.
ADO-2051939 UPF ISSU - some pods of other feds are evicted during the upgrade of fed-kube-addons	Currently it is not supported to change in fed-kube_addons values.yaml pod-node_config section values during upgrades of an application. In the current implementation of commissioning daemonset if there are parameter changes then it is required to perform api-server/controller manager restarts and/or kubelet restarts. Since these restarts are close to being simultaneous on all relevant nodes they can render k8s control plane disruptions. These disruptions may lead to unexpected pod evictions.

SMF

The following SMF anomalies are corrected in this release.

Table 24. Anomalies Corrected in SMF

Anomaly	Fix Description
ADO-1545663, IcM-498776504 When PFCP returns context not found during CHF Reauth Procedure SMF session retained and not deleted.	Corrected typo in variable name across multiple instances. A new condition is added to check if the process failed with context not found, and if so, the state is changed to Smfcc3gppProcDeactivateState. A flag is set to indicate that the process failed with context not found when the UPF session is not active.
ADO-1775614 Roaming (hSMF/vSMF) should work with MTLS enabled (https)	Update code to add support for Mutual TLS (MTLS) in roaming scenarios for hSMF/vSMF, ensuring compatibility with both HTTP and HTTPS protocols.

UPF

The following UPF anomalies are corrected in this release.

Table 25. Anomalies Corrected in UPF

Anomaly	Fix Description
ADO-1638224 – Clone, IcM-510425501 Change in the UCD server to drop session records with PDRs/QRs/URRs/FARs that exceeding a threshold	Added a mechanism to check if a Pfcpc message is too large and if so, avoid updating it in UCD. This is done to prevent potential issues with handling large Pfcpc messages.
ADO-1952830 – Clone UPF stats: PPE is pushing wrong NC name for the N6IMS IPv6 phy and sub-interface (i.e. IPv4-VRF:0) and this is misleading while querying stats on prom and grafana	Adjusted logic to correctly handle IPv6 sub-interface names when no IPv4 address is configured. Updated to skip pushing default network context names (e.g., ipv4-VRF:0, ipv6-VRF:0) to Prometheus to avoid double counting.

Anomaly	Fix Description
ADO-1964409 Nexus-VO: UPF Resiliency - 11K session loss after hard rebooting of worker node that hosts l3rm, ippe, redis-cluster master, upf-monitor	This is expected session loss as per design. Most of the resiliency features are fixed part of ISSU and this addresses essential gaps present in the UPF. We addressed the single point of failures to address session loss. In a few cases, we address multiple failures at the same time. This bug is due to 2 critical services running on same worker node and both down the same time due to rebooting the node. The 2 most critical services having issues at the same time due to software is very rare. So, fundamentally we need not handle this as it is not going to happen. This is possible in manual case only where worker node crashing having both services. The UPF can be deployed in such a way that n+1 worker nodes and make sure L3RM runs on node exclusively can avoid this case. This is purely deployment options. This is designed functional thinking L3RM and PPE runs on same node which reboots is avoidable. Reason for session loss: In general, the PPE holds n number of sessions as a primary and has another PPE as secondary for all these n sessions. When Primary PPE goes down, L3RM is the one going to update to secondary PPE pushing it to act as primary for all these n number of sessions. This bug sees that L3RM is missing which is ideally informs to secondary PPE and all the active sessions on primary PPE is down and there is no intimation to secondary PPE to own them. So, all these sessions ideally in a zombie state are lost eventually.
ADO-2020086 Nexus-VO: During the IPPE scale-down to 0 and back to 10, a VR was observed on the fed-UPF	There is a possibility of VR when PPE instances are scaled down to 0 and then scaled up to normal instances count. System (fed-upf) recovers on its own without intervention. Currently there are no plans to address this issue.

NRF

The following NRF anomalies are corrected in this release.

Table 26. Anomalies Corrected in NRF

Anomaly	Fix Description
ADO-2032777 - Clone, lCM-545247406 NRF should not return NFs in maintenance mode	Modified a method to check for undiscoverable status in addition to suspended status.

Anomalies fixed in UC5.2.0.x

N/A until the patch stream begins.